The Coherence-Delay Trade-off in Tracking Under Drift

A Cube-Root Law for Finite-Memory Estimation

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Abstract

Tracking in a changing environment is limited by a simple tension: more memory reduces noise, but it also makes the estimate older. In passive streams this is already a statistical problem; in closed-loop systems it becomes an epistemic one, because the same actions used to stabilize behavior can also make a stale model look correct. This paper treats those two points as a single story.

First, we formalize a coherence-delay trade-off for finite-memory tracking under drift. For the finite-memory averaging class studied here, the variance term falls with memory while the lag term grows with drift and effective age, yielding a **cube-root law**: the **minimum achievable tracking error** scales as $\boldsymbol{\zeta^{1/3}}$ with drift rate ζ, and the **optimal effective memory** scales as $\boldsymbol{\zeta^{-2/3}}$. Empirically, we recover slopes of 0.316 for sliding windows and 0.317 for EWMA, closely matching the theoretical prediction.

Second, we study the regime in which low observed error is maintained not by adaptation but by steering the environment toward a stale model. We call it coercive masking and introduce the **Coherence Index (CI)** and the **Effort-Adjusted Coherence Index** (CI^E) to separate genuine adaptation from action-induced alignment. On passive benchmarks, generic changepoint detectors remain the stronger default alarms. On active synthetic and logged recommendation benchmarks, however, CI^E exposes masking that passive detectors do not encode under the same calibration, identifying masking onset with median delay 5.0 steps in the particle benchmark and improving KuaiRand bubble detection from 0.458 to 0.692.

The resulting picture is narrow but useful: under drift, finite memory imposes a real error floor, and under feedback, raw fit can be misleading unless the cost of intervention is made explicit.

Keywords: tracking under drift; finite-memory estimation; Wasserstein distance; Sinkhorn divergence; concept drift; online compression; Coherence Index; Effort-Adjusted Coherence Index; coercive masking.
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1 Introduction

Modern adaptive systems do not observe a fixed world. They estimate, predict, rank, and control while the environment keeps changing beneath them, and in many cases their own actions help shape what they will observe next. In recommender systems, adaptive control, and streaming prediction, the problem is therefore not only estimation under noise. It is estimation under drift, with finite memory, and often under feedback.

That combination creates two distinct difficulties. The first is statistical. A finite-memory estimator improves when it can average more data, but every additional observation also comes from a more distant past. Under drift, memory reduces variance and increases staleness at the same time. The second is epistemic. In action-coupled systems, low observed error need not mean the model is current, because the system may be pushing the environment toward the model rather than updating the model to match the environment. A tracker can look coherent for the wrong reason.

This paper studies those two difficulties as parts of the same story. We first ask what finite memory can achieve when the target itself is moving. We then ask how to diagnose the special failure mode in which apparent coherence is being maintained by intervention rather than by adaptation.

The first answer is a structural coherence-delay trade-off. When observations are aggregated over time, total tracking error decomposes into a variance term that falls with memory length and a lag term that rises with drift and with the effective age of the data. Optimizing that balance yields a cube-root law for the finite-memory averaging class analyzed here: $\mathcal{E}_{\min} = \frac{3}{2} C_K^{2/3} \zeta^{1/3}$ with $n^* = (C_K/\zeta)^{2/3}$. This law is not presented as a universal theorem for all adaptive estimators. It is exact for the averaging class studied here, and in the critical-window regime the same exponent reappears as a minimax lower bound for arbitrary estimators restricted to the last m samples of a ζ-Lipschitz drift path. In that sense, tracking under drift is limited by the same object that makes online compression useful: memory itself.

The second answer concerns feedback masking. In closed-loop settings, good fit can be purchased by control effort. The system may appear well aligned with its environment because its actions are holding the environment near a stale model. In the synthetic control regime studied below, we call this *coercive masking*. To analyze it, we separate three functional roles: a representation of the world ($\mathcal{P}$), an actuation rule ($\mathcal{A}$), and an adaptation operator (Φ). This is a modeling decomposition for action-effective tracking systems, not a claim about every information process.

That decomposition yields two diagnostics. The Coherence Index (CI), denoted by $\mathcal{C}$, measures the apparent bottleneck of the update loop. The Effort-Adjusted Coherence Index (CI^E), denoted by $\mathcal{C}^E$, discounts alignment that is purchased through intervention rather than through model adaptation. The point is not to replace generic changepoint detectors on passive streams. On passive benchmarks, such detectors remain the stronger default alarms. The point is to expose a failure mode that passive alarms do not encode: the regime in which the world looks stable because the system is forcing it to look stable.

The paper proceeds in that order. We begin by fixing the minimal formal objects needed to talk about action-effective tracking, then show why closed-loop updates must account for present state, recent action, and new signal. With that structure in place, Section 6 derives the coherence-delay trade-off and the finite-memory cube-root law. The coherence diagnostics and their effort-aware variant are then introduced, and Section 7 tests both parts of the story: passive benchmarks situate the finite-memory result against generic drift alarms, while active and logged benchmarks test whether effort-aware coherence can surface coercive masking.

The scope is intentionally narrower than the headline might tempt. The cube-root law is exact for the finite-memory averaging class analyzed here, not for every recursive or extrapolative architecture. The decomposition is motivated only for *action-effective* systems; passive observers are a genuine boundary case. The stability results remain conditional on explicit ISS hypotheses, with the Gaussian tracker serving as the constructive reference instance. Within that scope, however, the picture is sharp: temporal staleness imposes a real tracking floor, and under feedback, raw fit is not yet a safe diagnostic.

2 Mathematical Definitions

We start with the smallest amount of formal structure needed for the rest of the paper: a state space, an information state, and the three operators that separate representation, actuation, and adaptation.

2.1 State Spaces and Information State

Definition 2.1 (State space)**.** Let S be a non-empty measurable space (the *signal space*). The *information state space* $\mathcal{I}$ is the space of all probability measures on S equipped with a metric $d_{\mathcal{I}}$ compatible with weak convergence. Unless otherwise specified, $d_{\mathcal{I}}$ denotes the L^2-Wasserstein distance W_2.

Definition 2.2 (Potentiality)**.** The *Potentiality* at time t is a measurable map

$$\mathcal{P}(S,t) \in \mathcal{I},$$

representing the system's current probability measure over possible world states. $\mathcal{P}$ is ε*-coherent* at time t if $W_2(\mathcal{P}(S,t),\, \mathcal{P}^*(S,t)) \le \varepsilon$, where $\mathcal{P}^*(S,t)$ denotes the true marginal distribution of the environment at time t.

Remark 2.3 (Computational realization of W_2). The L^2-Wasserstein distance W_2 is chosen as the metric on $\mathcal{I}$ for its geometric faithfulness: unlike f-divergences such as KL divergence, W_2 inherits the geometry of the underlying signal space S and behaves well under spatial translations of the distribution support, a desirable property when concept drift shifts the support of $\mathcal{P}^*$. In high-dimensional signal spaces, exact computation of W_2 costs $O(n^3 \log n)$ via network-flow solvers. For streaming applications requiring real-time Φ_t updates, the entropic regularization of Cuturi [2013] replaces W_2 with the Sinkhorn divergence S_ε, computable in $O(n^2/\varepsilon^2)$ via parallelizable matrix-scaling iterations [Altschuler et al., 2017]. S_ε converges to W_2^2 as $\varepsilon \to 0$ and breaks the curse of dimensionality of raw W_2 at the cost of a bias proportional to ε [Genevay et al., 2019]. The theoretical results hold for W_2; their computational realization in streaming contexts is demonstrated in the Gaussian-tracker experiments (section 7; code at appendix A).

Definition 2.4 (Actuation)**.** The *Actuation operator* is a measurable map

$$\mathcal{A}_t \colon \mathcal{I} \to \mathcal{O},$$

where $\mathcal{O}$ is the output space (distinct from lowercase o, which denotes a specific output element $o \in \mathcal{O}$). $\mathcal{A}_t$ is *δ-reproducible* if for any $p \in \mathcal{I}$ and any two executions ω, ω', $d_{\mathcal{O}}(\mathcal{A}_t(p;\omega), \mathcal{A}_t(p;\omega')) \le \delta$. A deterministic actuation is 0-reproducible.

Definition 2.5 (Convergence operator)**.** The *Convergence operator* is a measurable map

$$\Phi_t \colon \mathcal{I} \times \mathcal{O} \times S \to \mathcal{I},$$

mapping the current information state, the most recent actuation, and an incoming signal to an updated information state. Φ_t is *coherence-restoring* if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $t \ge 0$,

$$W_2(\mathcal{P}(S,t), \mathcal{P}^*(S,t)) \;\le\; \beta(W_2(\mathcal{P}(S,0), \mathcal{P}^*(S,0))\,, t) \;+ \gamma\left(\sup_{s \le t} \|u(s)\|\right),$$

where $u(s)$ represents the environmental drift input at time s.

Definition 2.6 (Three-Role System)**.** An information system $I(S,t)$ is a *three-role system* if it is equipped with all three operators $(\mathcal{P}, \mathcal{A}, \Phi)$ satisfying definitions 2.2, 2.4 and 2.5 respectively. The composite information state is

$$I(S,t) \coloneqq \big(\mathcal{P}(S,t), \mathcal{A}_t, \Phi_t\big) \in \mathcal{I} \times \mathcal{F}_{\mathcal{O}} \times \mathcal{F}_{\mathcal{I}}.$$

2.2 ISS Definitions

Definition 2.7 (Class $\mathcal{K}$, $\mathcal{KL}$)**.** A function $\gamma \colon \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ is class $\mathcal{K}$ if it is continuous, strictly increasing, and $\gamma(0) = 0$. A function $\beta \colon \mathbb{R}^2_{\ge 0} \to \mathbb{R}_{\ge 0}$ is class $\mathcal{KL}$ if $\beta(\cdot, t) \in \mathcal{K}$ for each t and $\beta(r, \cdot) \to 0$ as $t \to \infty$ for each $r > 0$.

Definition 2.8 (Input-to-State Stability)**.** A dynamical system $\dot{x} = f(x,u)$ is *input-to-state stable* (ISS) if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for every initial state $x(0)$, every essentially bounded input u, and all $t \ge 0$,

$$\|x(t)\| \le \beta(\|x(0)\|\,, t) + \gamma(\|u\|_\infty).$$

3 Why Action Enters the Update in Closed-Loop Tracking

The next step is to justify the minimal closed-loop architecture used throughout the paper. The finite-memory law studied later concerns tracking error once a representation is already in place, but in action-effective systems there is a prior question: what must an update rule remember in order to remain coherent under feedback? In open-loop settings, combining past state and new observations is enough. In closed-loop settings, however, the system's own actions alter future observations. The update therefore needs to keep track of what the system did, not just of what it saw.

3.1 External Characterization: Drift-Coherence

Definition 3.1 (Drift-Coherence)**.** An information system $I(S,t)$ is *drift-coherent* if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $t \geq 0$ and all environmental inputs with $\operatorname{ess\,sup}_{s \leq t} \|u(s)\| \leq M < \infty$:

$$W_2(\mathcal{P}(S,t),\, \mathcal{P}^*(S,t)) \;\leq\; \beta(W_2(\mathcal{P}(S,0),\, \mathcal{P}^*(S,0))\,,\, t) \;+\; \gamma(M).$$

The system maintains bounded tracking error under bounded drift.

Definition 3.2 (Action-Effectiveness)**.** The actuation operator $\mathcal{A}_t$ is *action-effective* in environment $\mathcal{X}$ if there exist distinct outputs $o, o' \in \mathcal{O}$ and a state $s \in S$ such that

$$D_{\mathrm{TV}}\Big(P(\cdot \mid s_t = s,\, o) \,\Big\|\, P(\cdot \mid s_t = s,\, o')\Big) \;>\; 0.$$

If this never occurs, the system is *passive*: its actions do not affect the environmental transition law.

3.2 Belief Update With Feedback

Lemma 3.3 (Belief Update Lemma)**.** *Let $I(S,t)$ be a drift-coherent, action-effective information system. Any update map U whose repeated application preserves drift-coherence in this class must depend on three arguments:*

(i) *the current internal state $p = \mathcal{P}(S,t)$,*

(ii) *the most recent action $o = \mathcal{A}_t(\mathcal{P})$, and*

(iii) *the incoming environmental signal s_{t+1}.*

Each argument cannot be dropped without leaving the class of drift-coherent updates considered here.

Proof. If U omits p, the update becomes memoryless and discards the sufficient summary of the past; under temporally structured drift this prevents bounded tracking error. If U omits o, then in an action-effective environment the update cannot distinguish exogenous drift from changes induced by the system's own actuation. This is precisely the credit-assignment problem that appears in POMDP filtering and belief update semantics [Kaelbling et al., 1998, Katsuno and Mendelzon, 1992, Bonanno, 2024]. If U omits s_{t+1}, no new environmental evidence enters the state estimate and the gap between $\mathcal{P}$ and $\mathcal{P}^*$ grows with drift. In each case, the drift-coherence bound fails. □

Remark 3.4 (Passive observer systems)*.* When actions do not influence transitions, the update can drop the actuation argument and the system is genuinely dyadic. The three-role decomposition is therefore motivated only for action-effective settings.

Corollary 3.5 (Ternary Update Structure)**.** *For action-effective tracking systems, any drift-coherent update map has the form*

$$U\colon \mathcal{I} \times \mathcal{O} \times S \to \mathcal{I}.$$

Any ostensibly smaller description either hides one of these arguments inside a state variable or fails drift-coherence.

Proof. This is immediate from lemma 3.3. □

Theorem 3.6 (Minimal Functional Roles). *Let $I(S,t)$ be a drift-coherent, action-effective information system. Any functional decomposition that preserves corollary 3.5 in this class must realize three roles: a stored state, an action channel, and an adaptation rule that maps the state, action, and new observation to an updated state. The tuple $(\mathcal{P}, \mathcal{A}, \Phi)$ is the explicit decomposition used here.*

Proof. By corollary 3.5, the update requires all three arguments (p, o, s_{t+1}). A decomposition that omits any one of the corresponding roles must either re-encode it implicitly inside another component or lose the update structure required for drift-coherence. The explicit realization used here is $\mathcal{P}$ for stored state, $\mathcal{A}$ for the action channel, and Φ for the update map. □

This necessity result does not yet tell us when such a three-role system is stable or diagnostically meaningful. It only shows that once actions affect future observations, a dyadic state-observation update is too small. The next section supplies the corresponding stability template under which the three roles can be scored.

4 A Minimal Stability Template for Closed-Loop Tracking

The necessity argument in section 3 identifies the three inputs a closed-loop update must use. The present section addresses the complementary question of sufficiency: under what conditions does that three-role decomposition support a meaningful diagnostic rather than just a formal one? To make the later scores interpretable, we need a stability template under which representation, actuation, and adaptation each admit a Lyapunov-style quantity. We use an ISS-based cascade model for that purpose.

4.1 Hypotheses

The following assumptions are explicit and application-dependent.

(H-ISS-P) The representation subsystem admits an ISS-Lyapunov function $V_P : \mathcal{I} \to \mathbb{R}_{\geq 0}$.

(H-ISS-A) The actuation subsystem admits an ISS-Lyapunov function V_A.

(H-ISS-Φ) The adaptation subsystem admits an ISS-Lyapunov function V_Φ.

(H-ISS-Env) The environmental drift input is essentially bounded and evolves slowly enough for Φ_t to remain contractive on the induced state trajectory.

Remark 4.1 (Measure-valued state spaces). When $\mathcal{P}(S,t)$ is measure-valued, the natural potential function is $V_P = \frac{1}{2}W_2(\mathcal{P}(S,t), \mathcal{P}^*(S,t))^2$. In Gaussian instances this reduces to an ordinary quadratic tracking error; in more general settings the construction follows the Wasserstein gradient-flow viewpoint [Ambrosio et al., 2005, Mironchenko and Prieur, 2020].

4.2 Main Stability Template

Theorem 4.2 (Conditional Borromean Stability Template)**.** *Under hypotheses* (H-ISS-P)*,* (H-ISS-A)*,* (H-ISS-Φ)*, and* (H-ISS-Env)*, the composite system* $I(S,t) = (\mathcal{P}, \mathcal{A}, \Phi)$ *with feedback structure*

$$\Phi_t \xrightarrow{\text{updates}} \mathcal{P}(S,t) \xrightarrow{\text{drives}} \mathcal{A}_t \xrightarrow{\text{feeds back}} \Phi_t$$

admits the composite Lyapunov functional $V_{\text{total}} := V_P + V_A + V_\Phi$*, which is non-increasing in expectation along trajectories for bounded environmental inputs. Removing any one of the three roles destroys this guarantee in general.*

Proof. The cascade ($\Phi \to \mathcal{P} \to \mathcal{A}$) is ISS by repeated application of standard cascade arguments [Sontag, 1989, 1995, Dashkovskiy et al., 2010]. The subsystem Lyapunov functions combine into a composite functional whose cross terms are absorbed by the ISS gains. If any one of the three roles is removed, at least one of the required input channels disappears: without Φ the state is not corrected, without $\mathcal{P}$ the action has no bounded state input, and without $\mathcal{A}$ the feedback loop to adaptation is broken. In each case the monotone decrease of V_{total} is no longer guaranteed. □

Remark 4.3 (Scope of the stability template)*.* Theorem 4.2 is a conditional stability template, not a universal theorem about all three-role architectures. Its force is constructive in cases where the subsystem Lyapunov functions can be written down and their gains checked directly. In this paper that verification is carried out only for the Gaussian tracker. For non-linear or black-box systems, the template should be read as a checklist of sufficient conditions rather than as an already verified property of the architecture itself.

Example 4.4 (Concrete Gaussian instantiation)**.** Let $\mathcal{P}^*(S,t) \sim \mathcal{N}(\mu_t, \Sigma)$ with

$$\mu_t = \mu_0 + \int_0^t \eta(s)\,ds, \qquad \|\eta\|_\infty < M.$$

Let $\mathcal{P}(S,t) = \mathcal{N}(\hat{\mu}_t, \Sigma)$ be the Kalman estimate, $\mathcal{A}_t(\mathcal{P}) = \hat{\mu}_t$, and Φ_t the Kalman update. Then $V_P = W_2(\mathcal{P}, \mathcal{P}^*)^2 = \|\hat{\mu}_t - \mu_t\|^2$, together with the standard actuation and innovation terms, yields a concrete instance satisfying the ISS template [Jazwinski, 1970]. This Gaussian case is a constructive reference instance for theorem 4.2.

With the stability template in place, the remaining question is diagnostic: what kinds of breakdown this decomposition is meant to reveal, and how those breakdowns should appear in the component scores. The next section names the failure modes that organize the empirical part of the paper.

5 Diagnostic Failure Modes

The decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ becomes useful only when it separates failure regimes that matter empirically. This section names the three failure modes that recur throughout the experiments and later feed directly into the coherence scores. The point is not to enumerate every possible pathology of an adaptive system, but to isolate the three ways in which representation, actuation, and adaptation can fail in the diagnostic story developed here.

Proposition 5.1 (Failure Mode 1: Static-$\mathcal{P}$ collapse)**.** *Suppose Φ_t ceases to update $\mathcal{P}$, so $\mathcal{P}(S,t) = \mathcal{P}(S,t_0)$ for all $t > t_0$. If the environment undergoes distributional shift $\mathcal{P}^*(S,t) \neq \mathcal{P}^*(S,t_0)$ for large t, then $W_2(\mathcal{P}(S,t), \mathcal{P}^*(S,t)) \to \infty$ and the expected actuation error $\mathbb{E}[d_{\mathcal{O}}(\mathcal{A}_t(\mathcal{P}), \mathcal{A}_t^*)]$ is unbounded.*

Proof. $\mathcal{P}(S,t) = \mathcal{P}(S,t_0)$ is fixed while $\mathcal{P}^*$ drifts, so $W_2 \to \infty$ by definition. Lipschitz continuity of $\mathcal{A}_t$ then propagates the divergence to actuation error. □

Remark 5.2 (Interpretation)*.* FM-1 is the stale-model regime: the representation no longer tracks the world, so downstream actions inherit an increasingly obsolete state estimate.

Proposition 5.3 (Failure Mode 2: Non-deterministic-$\mathcal{A}$ divergence)**.** *If $\mathcal{A}_t$ is not δ-reproducible for any $\delta < \infty$, then $H(\mathcal{A}_t \mid \mathcal{P}(S,t)) > 0$ for all t: actuation carries information not determined by the state model. Distributed replicas of the system diverge, and the composite resists auditing and replication.*

Proof. Non-δ-reproducibility implies $\exists\, \omega \neq \omega'$ with $\mathcal{A}_t(p;\omega) \neq \mathcal{A}_t(p;\omega')$, giving positive conditional entropy. Auditability requires reconstructing actuation from state; this fails when conditional entropy is positive. □

Remark 5.4 (Interpretation)*.* FM-2 is the non-reproducible-action regime: the model may still be accurate, but the realized action can no longer be reconstructed from the represented state, so replicas and audits diverge.

Proposition 5.5 (Failure Mode 3: Absent-Φ monopolization)**.** *Let $\Phi_t \equiv \mathrm{id}$ (no update). In any closed-loop deployment, actuation feedback shifts $\mathcal{P}^*$; since $\mathcal{P}$ cannot track, the distributional mismatch grows at a rate lower-bounded by the rate of environmental response to actuation.*

Proof. The closed-loop case follows from $\mathcal{P}^*(t+1) \leftarrow f(\mathcal{P}^*(t), \mathcal{A}_t(\mathcal{P}(t)))$ with $\mathcal{P}$ fixed, so the gap $W_2(\mathcal{P}, \mathcal{P}^*)$ grows at least as fast as the variation of f in its second argument whenever $\mathcal{A}_t(\mathcal{P}) \neq \mathcal{A}_t^*$. □

Remark 5.6 (Interpretation)*.* FM-3 is the missing-adaptation regime. In passive settings it appears as a failure to track external drift; in closed-loop settings it can amplify the system's own feedback and produce self-reinforcing misalignment.

These failure modes motivate the score construction in the next section. Once the three ways of failing are separated, the natural diagnostic question is how to summarize them in a single quantity without losing the identity of the active bottleneck.

6 Coherence Diagnostics

The coherence-delay law constrains the representation term of the finite-memory averaging systems studied here, but closed-loop performance also depends on whether actions remain reproducible and whether adaptation remains effective. The ISS template of section 4 provides one way to score those three ingredients. This section packages them into a diagnostic built from the minimal decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ used throughout the paper.

Definition 6.1 (Component Coherence Scores)**.** Let $I(S,t) = (\mathcal{P}, \mathcal{A}, \Phi)$ be an information system with ISS-Lyapunov functionals V_P, V_A, V_Φ as in section 4. Define the three *component coherence scores* at time t as:

$$\sigma_P(t) \coloneqq \frac{1}{1+V_P(t)} = \frac{1}{1+\frac{1}{2}W_2(\mathcal{P}(S,t),\, \mathcal{P}^*(S,t))^2}, \tag{1}$$

$$\sigma_A(t) \coloneqq \exp(-H(\mathcal{A}_t \mid \mathcal{P}(S,t)))\,, \tag{2}$$

$$\sigma_\Phi(t) \coloneqq \frac{1}{1+V_\Phi(t)}. \tag{3}$$

Each score takes values in $(0,1]$, equals 1 when the corresponding component is at its reference state, and approaches 0 as the component degrades.

Definition 6.2 (Coherence Index)**.** The *Coherence Index (CI)* of a system $I(S,t)$ at time t is

$$\mathcal{C}(I(S,t),t) \;\coloneqq\; \min(\sigma_P(t),\; \sigma_A(t),\; \sigma_\Phi(t)) \;\in\; (0,1].$$

The min aggregation is the correct choice given the Borromean property (theorem 4.2): system coherence is bounded by the weakest component, and no strength in two components compensates for failure in the third. An arithmetic mean would obscure localized failures; the min preserves the diagnostic identity of the bottleneck.

Remark 6.3 (Bottleneck sensitivity and effort correction)*.* The use of $\min(\sigma_P, \sigma_A, \sigma_\Phi)$ is bottleneck-sensitive: it diagnoses the weakest active subsystem, not overall system quality in the round. Aggregators such as arithmetic or geometric means may be reported as descriptive summaries, but they are not valid replacements for the structural diagnostic given by definition 6.2.

In action-effective environments, however, high actuation capacity can keep σ_P artificially high by forcing the world toward a stale model. When such feedback masking is possible, a conservative extension is the *effort-corrected score*

$$\mathcal{C}^E(I(S,t),t) \;\coloneqq\; \min\!\big(\sigma_P(t)e^{-\lambda E_t/E_0},\; \sigma_A(t),\; \sigma_\Phi(t)\big)\,,$$

where E_t is any monotone proxy for corrective actuation burden, $E_0 > 0$ is a reference scale, and $\lambda > 0$ controls sensitivity. It discounts apparent improvements in model-world alignment when those gains come from action-induced confounding rather than better state estimation. In the synthetic particle tracker, E_t is the realized coercive effort $\alpha|a_t - s_t^{\mathrm{free}}|$; in deployed systems, action increments, energy expenditure, or control-surface usage can play the same role. For logged recommendation benchmarks, the preferred instantiation is the per-user KL divergence between the current exposure-tag distribution and the healthy random-exposure baseline distribution, aggregated sequentially over time and then batched across users.

Remark 6.4 (Finite-memory interpretation)*.* The scaling law behind proposition 6.9 is not an artifact of the sliding-window estimator alone. The two terms in the bound, $n^{-1/2}$ and $\zeta\tau$, encode the structural trade-off between variance reduction and delay-induced bias for the finite-memory averaging family studied here: the first term reflects the statistical limit of estimating a distribution from finitely many samples under subgaussian noise, and the second reflects the mismatch induced by drift when the estimator necessarily looks at stale information.

Concretely, the paper proves the cube-root scaling for the canonical local averaging schemes used throughout the experiments, namely sliding windows and exponentially weighted averages. The same exponent reappears because these estimators expose the same variance-vs.-lag balance under drift. Below we also show a minimax lower bound for arbitrary measurable estimators that only see the last m samples of a ζ-Lipschitz drift path. We do not claim here a universal minimax theorem over arbitrary recursive, trend-extrapolating, or otherwise unconstrained adaptive estimators.

Within the averaging class studied here, the $\zeta^{1/3}$ law should be read as a robust operational regime under the stated regularity assumptions rather than as a peculiarity of a single windowing scheme.

6.1 Estimable Score

Definition 6.1 requires $\mathcal{P}^*(S,t)$, the true oracle distribution of the environment, to compute $\sigma_P(t)$. In any deployed system, $\mathcal{P}^*$ is unobservable by construction: a system with complete knowledge of $\mathcal{P}^*$ would have nothing to estimate. We therefore introduce an estimable version of σ_P using an empirical approximation of $\mathcal{P}^*$.

Definition 6.5 (Empirical Potentiality Estimator)**.** Let $\{s_{t-n+1},\dots,s_t\}$ be the n most recent environmental signals observed by the system. The *empirical potentiality estimator* is the empirical measure:

$$\hat{\mathcal{P}}^*(S,t) \;\coloneqq\; \frac{1}{n}\sum_{k=t-n+1}^{t}\delta_{s_k},$$

where δ_{s_k} is the Dirac measure at observation s_k.

Definition 6.6 (Estimable Score)**.** The *debiased Sinkhorn divergence* between measures μ, ν is

$$S_\varepsilon(\mu,\nu) \;\coloneqq\; T_\varepsilon(\mu,\nu) - \tfrac{1}{2}T_\varepsilon(\mu,\mu) - \tfrac{1}{2}T_\varepsilon(\nu,\nu),$$

where T_ε is the raw entropically regularized optimal transport cost. Unlike T_ε, the debiased divergence satisfies $S_\varepsilon(\mu,\mu)=0$, restoring the metric property and enabling the dimension-independent convergence rate of proposition 6.9.

The *estimable component* $\hat{\sigma}_P$ replaces $\mathcal{P}^*$ with $\hat{\mathcal{P}}^*$ in eq. (1) using S_ε:

$$\hat{\sigma}_P(t) \;\coloneqq\; \frac{1}{1+\frac{1}{2}S_\varepsilon\Big(\mathcal{P}(S,t),\,\hat{\mathcal{P}}^*(S,t)\Big)^2}.$$

The *estimable score* is

$$\widehat{\mathcal{C}}(I(S,t),t) \;\coloneqq\; \min(\hat{\sigma}_P(t),\;\sigma_A(t),\;\sigma_\Phi(t))\,.$$

The three components have different operational meanings. In the Gaussian tracker, σ_A is exact because the action channel is deterministic. In the particle tracker, σ_A is a controlled synthetic surrogate based on the actuation gap. In KuaiRand, σ_A is a logged-data surrogate based on normalized exposure entropy, available only because the benchmark exposes the necessary exposure counts. Likewise, $\sigma_\Phi = 1/(1+V_\Phi)$ is exact only when the convergence subsystem's Lyapunov functional is available or constructively specified.

We therefore treat action observability as a three-level ladder: 'exact' when the action channel is deterministic or constructively known, 'identified' when the channel is logged richly enough to estimate the target quantity with a direct estimator, and 'proxy' when the benchmark only supports an operational surrogate. The Gaussian tracker sits at the exact level, the synthetic particle benchmark sits at the identified/surrogate boundary, and KuaiRand sits at the proxy level.

Assumption 6.7 (W_2-Lipschitz Drift)**.** The oracle trajectory $t \mapsto \mathcal{P}^*(S,t)$ is W_2-Lipschitz with drift rate $\zeta \geq 0$: for all $s, t \geq 0$,

$$W_2(\mathcal{P}^*(S,s),\, \mathcal{P}^*(S,t)) \;\leq\; \zeta\,|t-s|.$$

Remark 6.8 (Verification in the Gaussian tracker)*.* For the Gaussian family $\mathcal{P}^*(S,t) = \mathcal{N}(\mu^*(t), \Sigma)$ with common covariance Σ, the Wasserstein distance reduces to translation of the mean:

$$W_2(\mathcal{N}(\mu_s,\Sigma), \mathcal{N}(\mu_t,\Sigma)) = \|\mu_s - \mu_t\|_2.$$

Hence the Gaussian tracker satisfies assumption 6.7 whenever $\|\dot{\mu}^*\|_\infty \leq \zeta$.

Proposition 6.9 (Estimation Error Bound for $\hat{\sigma}_P$)**.** *Let assumption 6.7 hold, and assume each oracle marginal $\mathcal{P}^*(S,t)$ is K-subgaussian. Let $\hat{\mathcal{P}}^*$ be the sliding-window empirical estimator with window n (definition 6.5), and let $\tau = (n-1)/2$ be the effective lag of the centered window. Then under the entropic regularization of remark 2.3,*

$$\mathbb{E}\left|\sigma_P(t) - \hat{\sigma}_P(t)\right| \;\leq\; C_K\, n^{-1/2} + \zeta\tau,$$

where C_K absorbs the finite-sample Sinkhorn constant, the fixed regularization ε, and the uniform subgaussian envelope. This is a finite-memory upper bound for the sliding-window class, not a minimax statement over all adaptive estimators.

Proof. Let

$$\bar{\mathcal{P}}^*(S,t) \;:=\; \frac{1}{n}\sum_{k=t-n+1}^{t} \mathcal{P}^*(S,k)$$

be the oracle window-average law. The finite-sample term is the usual subgaussian Sinkhorn estimation error: by Genevay et al. [2019, Thm. 3], the debiased divergence between $\bar{\mathcal{P}}^*(S,t)$ and its empirical estimator is of order $n^{-1/2}$, with the regularization dependence absorbed into C_K.

The additional error comes from temporal staleness of the window. By convexity of W_2^2 in its second argument,

$$W_2\big(\mathcal{P}^*(S,t), \bar{\mathcal{P}}^*(S,t)\big) \leq \frac{1}{n}\sum_{k=t-n+1}^{t} W_2(\mathcal{P}^*(S,t), \mathcal{P}^*(S,k))\,.$$

Applying assumption 6.7,

$$\begin{aligned} W_2\big(\mathcal{P}^*(S,t), \bar{\mathcal{P}}^*(S,t)\big) &\leq \frac{\zeta}{n}\sum_{k=t-n+1}^{t} |t-k| \\ &= \zeta\,\frac{n-1}{2} = \zeta\tau. \end{aligned}$$

This is the explicit lag-bias term. Combining it with the $C_K n^{-1/2}$ statistical term yields the stated decomposition. Finally, the score map $x \mapsto 1/(1+\frac{1}{2}x^2)$ is globally Lipschitz on $\mathbb{R}_{\geq 0}$, so the same order transfers to $\mathbb{E}|\sigma_P(t) - \hat{\sigma}_P(t)|$. □

Remark 6.10 (Tightness of the lag-bias term). The lag term is not merely an artifact of upper-bounding. For monotone translation trajectories $\mathcal{P}^*(S,t) = \mathcal{P}_0(S - vt)$ with $\|v\| = \zeta$, the current law and the centered window-average law are separated by exactly the mean age of the window, so the temporal bias is of order $\zeta\tau$. The cube-root balance therefore reflects a real finite-memory limitation rather than a proof-only looseness.

proposition 6.9 grounds $\widehat{\mathcal{C}}$: the estimation error decreases as the window n grows and increases proportionally to drift rate ζ. The required window size for target error η is $n = O(C_{K,\varepsilon}^2/\eta^2)$, independent of dimension.

This characterization also yields the tracking limit used throughout the paper. When window lag is accounted for explicitly ($\tau = n/2$ for a centred window), the total estimation error decomposes as a sum of two opposing terms. Minimising over n yields the cube-root optimum for the averaging class, including the sliding-window and exponentially weighted schemes used in the paper.

Definition 6.11 (Lag-Inclusive Estimation Error)**.** For a sliding window of size n centred at lag $\tau = n/2$, the total expected estimation error is:

$$\mathcal{E}(n) \;=\; C_K\, n^{-1/2} \;+\; \tfrac{1}{2}\,\zeta\, n,$$

where the first term is the *variance* (decreasing in n) and the second is the *lag-bias* (increasing in n).

Proposition 6.12 (Optimal Window Under Drift)**.** *Under definition 6.11, the estimation error $\mathcal{E}(n)$ is minimised at the* optimal window

$$n^* \;=\; \left(\frac{C_K}{\zeta}\right)^{2/3},$$

yielding the minimum achievable tracking error

$$\mathcal{E}_{\min} \;=\; \tfrac{3}{2}\, C_K^{2/3}\, \zeta^{1/3}.$$

Proof. Setting $d\mathcal{E}/dn = 0$: $-\frac{1}{2}C_K n^{-3/2} + \frac{1}{2}\zeta = 0$, giving $n^* = (C_K/\zeta)^{2/3}$. Substituting back: $\mathcal{E}(n^*) = C_K(C_K/\zeta)^{-1/3} + \frac{1}{2}\zeta(C_K/\zeta)^{2/3} = C_K^{2/3}\zeta^{1/3} + \frac{1}{2}C_K^{2/3}\zeta^{1/3} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$. ☐

Corollary 6.13 (EWMA Analogue)**.** *Let the finite-memory estimator use exponential weights $w_j = \alpha(1-\alpha)^j$ for $j \geq 0$, so its effective lag is $\tau_\alpha = \sum_{j\geq 0} j w_j = (1-\alpha)/\alpha$ and its effective sample size is $n_{\mathrm{eff}}(\alpha) = 1/\sum_{j\geq 0} w_j^2 = (2-\alpha)/\alpha$. Under assumption 6.7, the resulting finite-memory error obeys*

$$\mathcal{E}_{\mathrm{EWMA}}(\alpha) \;\leq\; C_K\sqrt{\frac{\alpha}{2-\alpha}} + \zeta\,\frac{1-\alpha}{\alpha}.$$

Hence, in the small-α regime, $\alpha^* \propto (\zeta/C_K)^{2/3}$ and $\mathcal{E}_{\mathrm{EWMA,min}} \propto C_K^{2/3}\zeta^{1/3}$.

Proof. The variance term is the sliding-window rate with n replaced by the effective sample size $n_{\mathrm{eff}}(\alpha)$. The lag term is the W_2-Lipschitz drift rate multiplied by the effective age τ_α. For small α this becomes $\mathcal{E}_{\mathrm{EWMA}}(\alpha) = O(C_K\alpha^{1/2}+\zeta\alpha^{-1})$. Balancing the two exponents gives $\alpha^* = O((\zeta/C_K)^{2/3})$, and substitution yields the same $\zeta^{1/3}$ minimum-error law as the sliding-window case. ☐

Definition 6.14 (Causal Averaging Kernel)**.** A *causal averaging kernel* is a sequence of weights $w = (w_0, w_1, \dots)$ such that $w_j \geq 0$ and $\sum_{j\geq 0} w_j = 1$. Its effective size and effective lag are

$$n_{\text{eff}}(w) := \frac{1}{\sum_{j\geq 0} w_j^2}, \qquad \tau_{\text{eff}}(w) := \sum_{j\geq 0} j w_j.$$

Proposition 6.15 (Averaging-Kernel Lower Bound)**.** *Let $X_t = \mu_t + \xi_t$ with $\xi_t \sim \mathcal{N}(0, \sigma^2)$ i.i.d. and let the drift trajectory be the monotone translation model $\mu_t = \zeta t$ with $\zeta \geq 0$. For any causal averaging kernel w as in definition 6.14, the kernel estimator*

$$\hat{\mu}_t^{(w)} := \sum_{j\geq 0} w_j X_{t-j}$$

satisfies

$$\sqrt{\mathbb{E}\left(\hat{\mu}_t^{(w)} - \mu_t\right)^2} \;\geq\; c_1\, \sigma\, n_{\text{eff}}(w)^{-1/2} \vee c_2\, \zeta\, \tau_{\text{eff}}(w)$$

for universal constants $c_1, c_2 > 0$. Equivalently, the mean-squared error obeys the exact decomposition

$$\mathbb{E}\left(\hat{\mu}_t^{(w)} - \mu_t\right)^2 \;=\; \sigma^2 n_{\text{eff}}(w)^{-1} + \zeta^2 \tau_{\text{eff}}(w)^2,$$

and therefore the RMSE lower bound above. In particular, optimizing over the averaging family yields the same cube-root exponent as proposition 6.12.

Proof. The estimator error decomposes as

$$\hat{\mu}_t^{(w)} - \mu_t = \underbrace{\zeta \sum_{j\geq 0} j w_j}_{\text{bias}} + \underbrace{\sum_{j\geq 0} w_j \xi_{t-j}}_{\text{noise}}\,.$$

Hence the bias magnitude is exactly $\zeta \tau_{\text{eff}}(w)$ and the noise term is Gaussian with variance $\sigma^2 \sum_{j\geq 0} w_j^2 = \sigma^2 / n_{\text{eff}}(w)$. Therefore

$$\mathbb{E}\left(\hat{\mu}_t^{(w)} - \mu_t\right)^2 = \zeta^2 \tau_{\text{eff}}(w)^2 + \sigma^2 n_{\text{eff}}(w)^{-1},$$

and taking square roots gives the stated lower bound, since $\sqrt{a+b} \geq a^{1/2} \vee b^{1/2}$ for nonnegative a, b. Optimizing the resulting trade-off over the averaging family recovers the same 2/3 window scaling and 1/3 error exponent as the upper bound. $\square$

The kernel result above is exact but still tied to linear averaging. The next proposition removes that linearity assumption on the estimator side by moving to the minimax problem over all measurable rules that only observe a finite recent window. The price is that the signal class is now the full discrete ζ-Lipschitz family rather than the single monotone translation witness.

Proposition 6.16 (Window-Restricted Minimax Lower Bound)**.** *Fix an integer $m \geq 1$ and suppose the estimator only observes the last m samples*

$$X_{t-m+1}, \dots, X_t, \qquad X_s = \mu_s + \xi_s, \qquad \xi_s \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2).$$

Let $\mathcal{L}_m(\zeta)$ be the discrete ζ-Lipschitz class

$$\mathcal{L}_m(\zeta) := \{(\mu_{t-m+1}, \ldots, \mu_t) : |\mu_i - \mu_j| \le \zeta|i-j| \text{ for all } i, j\}\,.$$

For any measurable estimator $\hat{\mu}_t = \delta_m(X_{t-m+1:t})$, define the worst-case root risk

$$R_m(\delta_m) := \sup_{\mu \in \mathcal{L}_m(\zeta)} \Big(\mathbb{E}_\mu(\hat{\mu}_t - \mu_t)^2\Big)^{1/2}.$$

Then there exists a universal constant $c > 0$ such that

$$\inf_{\delta_m} R_m(\delta_m) \;\ge\; c \max\Big\{\sigma m^{-1/2},\; \sup_{1\le h\le m} \min\big(\sigma h^{-1/2},\; \zeta h\big)\Big\}\,.$$

In particular, if $1 \le (\sigma/\zeta)^{2/3} \le m$, then

$$\inf_{\delta_m} R_m(\delta_m) \;\ge\; c\,\sigma^{2/3}\zeta^{1/3}.$$

Proof. Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$.

We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in $\mathcal{L}_m(\zeta)$. For the two hypotheses $\theta_\pm = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\mathrm{KL}(P_+, P_-) = \frac{1}{2\sigma^2}\sum_{j=1}^{m}(\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}}\right)^2 = \frac{1}{2}.$$

Le Cam's two-point lemma therefore gives

$$R_m^* \;\ge\; c_0\,\sigma m^{-1/2}$$

for a universal constant $c_0 > 0$.

The second obstruction is local drift. Fix any $1 \le h \le m$ and define

$$\beta_h := \min\Big\{\zeta,\; \frac{\sigma}{4h^{3/2}}\Big\}, \qquad \mu_{t-j}^{\pm,h} := \pm\,\beta_h\,(h-j)_+, \qquad 0 \le j \le m-1,$$

where $(u)_+ := \max\{u, 0\}$. Each path belongs to $\mathcal{L}_m(\zeta)$ because its slope magnitude is at most $\beta_h \le \zeta$. The endpoint gap is

$$|\mu_t^{+,h} - \mu_t^{-,h}| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\mathrm{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2}\sum_{j=0}^{m-1}\big(2\beta_h(h-j)_+\big)^2 \le \frac{2\beta_h^2}{\sigma^2}\sum_{r=1}^{h} r^2 \le \frac{2\beta_h^2}{\sigma^2}h^3 \le \frac{1}{8}.$$

Applying Le Cam again yields

$$R_m^* \;\ge\; c_1\,\beta_h h \;\ge\; \frac{c_1}{4}\,\min\big\{\zeta h,\; \sigma h^{-1/2}\big\}$$

for another universal constant $c_1 > 0$. Since h was arbitrary,

$$R_m^* \;\geq\; \frac{c_1}{4} \sup_{1 \leq h \leq m} \min\bigl\{\zeta h,\ \sigma h^{-1/2}\bigr\}\,.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume $1 \leq h_\star := (\sigma/\zeta)^{2/3} \leq m$. Choose an integer $\bar{h} \in [h_\star/2, h_\star]$. Then

$$\zeta \bar{h} \geq \frac{1}{2}\,\zeta h_\star = \frac{1}{2}\,\sigma^{2/3}\zeta^{1/3},$$

and

$$\sigma \bar{h}^{-1/2} \geq \sigma h_\star^{-1/2} = \sigma^{2/3}\zeta^{1/3}.$$

Therefore

$$\min\bigl\{\sigma \bar{h}^{-1/2},\ \zeta \bar{h}\bigr\} \geq \frac{1}{2}\,\sigma^{2/3}\zeta^{1/3},$$

and the ramp bound gives

$$R_m^* \geq c\,\sigma^{2/3}\zeta^{1/3}$$

after absorbing constants. This proves the claim. □

Remark 6.17 (Interpretation: a finite-memory floor). Proposition 6.12 establishes that perfect coherence ($V_P = 0$) is *impossible* under continuous drift for the finite-memory averaging class analyzed here. Every drift-coherent system must carry a residual tracking error at least $\mathcal{E}_{\min} \propto \zeta^{1/3}$. This can be read as a finite-memory limitation: the two sources of error (variance and lag) trade against each other with no joint optimum of zero. The cube-root scaling has a concrete engineering consequence: a fifty-fold increase in drift rate ζ raises the minimum error floor by only $50^{1/3} \approx 3.7\times$, while simultaneously forcing the optimal window n^* to contract by $50^{2/3} \approx 14\times$. The optimal window $n^* = (C_K/\zeta)^{2/3}$ provides a *self-tuning rule*: given a real-time estimate $\hat{\zeta}_t$ of the local drift rate, the system should maintain window size $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$ to operate at the minimum of the error U-curve at every step. This turns the coherence-delay law into a practical self-tuning rule for the windowed estimators analyzed here. Proposition 6.16 shows that this is not only an artifact of linear averaging: over the broader ζ-Lipschitz drift class, any estimator restricted to the last m samples faces the same cube-root obstruction in the critical-window regime $1 \leq (\sigma/\zeta)^{2/3} \leq m$.

For small relative calibration errors, the rule is predictably sublinear:

$$\frac{\Delta n_t^*}{n_t^*} \approx \frac{2}{3}\left(\frac{\Delta C_K}{C_K} - \frac{\Delta \hat{\zeta}_t}{\hat{\zeta}_t}\right), \qquad \frac{\Delta \mathcal{E}_{\min}}{\mathcal{E}_{\min}} \approx \frac{2}{3}\frac{\Delta C_K}{C_K} + \frac{1}{3}\frac{\Delta \hat{\zeta}_t}{\hat{\zeta}_t}.$$

Thus moderate multiplicative misspecification in $\hat{\zeta}_t$ or C_K moves the operating point in the expected 2/3-power fashion rather than causing a linear blow-up. In practice, this implies that improvements in apparent tracking performance must be interpreted relative to this baseline, particularly in systems where actions influence observations.

Remark 6.18 (Practical estimation of V_Φ for black-box Φ_t)*.* The estimable score requires $\sigma_\Phi = 1/(1 + V_\Phi)$, where V_Φ is the ISS-Lyapunov functional for the convergence subsystem. When Φ_t is a black box, with internal dynamics unknown and ISS comparison functions $(\alpha_{1\Phi}, \alpha_{2\Phi}, \alpha_{3\Phi}, \chi_\Phi)$ unavailable, V_Φ admits no closed-form derivation. One possible route is Counterexample-Guided Inductive Synthesis (CEGIS), in which a learner proposes a candidate $\hat{V}_\Phi$ from observed state trajectories and a verifier attempts to falsify its dissipation inequality on sampled trajectories. We have not implemented that pipeline in the present codebase, so it should be read as future work rather than as a validated part of the current diagnostic. For opaque adaptation maps, the present paper should therefore be interpreted as providing the template requirement on V_Φ, not an operational estimator for every black-box Φ_t.

6.2 Architecture and Behavior Are Not Equivalent

A structural asymmetry in the framework deserves explicit statement.

Drift-coherence implies the need for all three roles. By lemma 3.3 and theorem 3.6, any drift-coherent, action-effective system must implement the three functional roles $(\mathcal{P}, \mathcal{A}, \Phi)$. Drift-coherence is the stronger behavioral condition; the decomposition is its architectural consequence in action-effective settings.

Implementing the decomposition is not enough. A system may implement $(\mathcal{P}, \mathcal{A}, \Phi)$ structurally while failing to satisfy the ISS hypotheses (H-ISS-P/A/Φ/Env) required for theorem 4.2. The components may be present but improperly tuned, have inadequate bandwidth, or lack sufficient expressiveness.

Proposition 6.19 (The Score Characterizes Component-Level Failure)**.** *Let $I(S,t)$ be a three-role system under hypotheses (H-ISS-P/A/Φ/Env). Then:*

(i) $\mathcal{C}(I(S,t),t) = 1$ *if and only if* $V_P(t) = 0$, $H(\mathcal{A}_t \mid \mathcal{P}) = 0$, *and* $V_\Phi(t) = 0$ *simultaneously.*

(ii) $\mathcal{C}(I(S,t),t) \to 0$ *if and only if at least one of the three failure modes is active:* $V_P(t) \to \infty$ (FM-1), $H(\mathcal{A}_t \mid \mathcal{P}) \to \infty$ (FM-2), *or* $V_\Phi(t) \to \infty$ (FM-3).

(iii) *The identity of* $\arg\min(\sigma_P, \sigma_A, \sigma_\Phi)$ *names the bottleneck component.*

Proof. (i) Each score is a strictly decreasing function of its Lyapunov functional or conditional entropy: $\sigma_P = 1 \Leftrightarrow V_P = 0$, $\sigma_A = 1 \Leftrightarrow H(\mathcal{A}_t \mid \mathcal{P}) = 0$, $\sigma_\Phi = 1 \Leftrightarrow V_\Phi = 0$. The min equals 1 iff all three equal 1 simultaneously.

(ii) As $V_P(t) \to \infty$, $\sigma_P(t) \to 0$, so $\mathcal{C} \to 0$; similarly for $V_\Phi(t) \to \infty$. As $H(\mathcal{A}_t \mid \mathcal{P}) \to \infty$, $\sigma_A(t) = e^{-H} \to 0$, so $\mathcal{C} \to 0$. The converse holds because $\mathcal{C} \to 0$ requires at least one $\sigma_i \to 0$, which by the above requires the corresponding functional to diverge.

(iii) $\mathcal{C} = \min_i \sigma_i$; the argmin identifies which subsystem's functional is largest, i.e., which component is furthest from its reference state. □

7 Empirical Validation

A Gaussian tracker provides an analytically tractable closed-loop tracking model that instantiates the three roles $(\mathcal{P}, \mathcal{A}, \Phi)$ and admits closed-form computation of all Lyapunov functionals.

The experiments verify the component-wise stability template by controlled ablation of each component under pure drift (section 7.2), validate the estimation error bound of proposition 6.9 numerically and characterize the lag-drift tradeoff (section 7.3), and demonstrate the score as a pre-failure early-warning diagnostic on a real concept-drift dataset (section 7.4). Code and reproduction instructions are provided in appendix A.

7.1 Formalization

The Gaussian tracker extends example 4.4. Let the environment produce observations from a drifting Gaussian:

$$s_t \sim \mathcal{N}\big(\mu^*(t),\, \sigma^2_{\text{obs}}\big)\,, \qquad \mu^*(t) = \mu_0 + \zeta t + \sigma_w B_t,$$

where B_t is standard Brownian motion. The tracker components are:

- $\mathcal{P}(S,t) = \mathcal{N}(\hat{\mu}_t, \hat{\sigma}_t^2)$: Gaussian belief state.
- $\mathcal{A}_t(\mathcal{P}) = \hat{\mu}_t$: deterministic mean extraction ($H(\mathcal{A}_t \mid \mathcal{P}) = 0$, $\delta = 0$).
- Φ_t: Kalman measurement update (see example 4.4).

All three Lyapunov functionals are available in closed form (example 4.4), making the tracker a convenient test bed for the trade-off and diagnostic claims studied here.

Lemma 7.1 (The Gaussian Tracker Satisfies the ISS Hypotheses). *The Gaussian tracker with $\sigma_w > 0$ and $\sigma_{\text{obs}} > 0$ satisfies all four ISS hypotheses (H-ISS-P/A/Φ/Env) of theorem 4.2:*

(i) ***H-ISS-P****: $V_P(t) = \frac{1}{2}W_2(\mathcal{P}, \mathcal{P}^*)^2 = \frac{1}{2}(\hat{\mu}_t - \mu^*(t))^2$ is a valid ISS-Lyapunov functional; the Kalman update contracts V_P at rate $1 - K_t$ per step, where $K_t \in (0,1)$ is the Kalman gain.*

(ii) ***H-ISS-A****: $V_A = H(\mathcal{A}_t|\mathcal{P}) = 0$ since $\mathcal{A}_t = \hat{\mu}_t$ is deterministic; H-ISS-A is satisfied trivially.*

(iii) ***H-ISS-Φ****: the Kalman recursion for $\hat{\sigma}_t^2$ converges to the steady-state variance $\hat{\sigma}^2_\infty = \frac{1}{2}(\sigma_w^2 + \sqrt{\sigma_w^4 + 4\sigma_w^2\sigma^2_{\text{obs}}})$, a stable fixed point; $V_\Phi = |\hat{\sigma}_t^2 - \hat{\sigma}^2_\infty|$ decays geometrically with rate $1 - K_\infty$.*

(iv) ***H-ISS-Env****: the environmental drift rate ζ is bounded by assumption; H-ISS-Env holds whenever $\zeta < (1 - K_\infty)\,\sigma_{\text{obs}}$.*

Proof. Items (i)–(iii) follow directly from the Kalman filter's convergence properties [Sontag, 1989, Mironchenko and Prieur, 2020]. Item (iv) is the H-ISS-Env condition of theorem 4.2 applied to the scalar Gaussian case; the contraction rate equals the steady-state Kalman gain $K_\infty = \hat{\sigma}^2_\infty/(\hat{\sigma}^2_\infty + \sigma^2_{\text{obs}})$. □

7.2 Experiment A: Component Ablation

Protocol. We run $T = 500$ steps with $\zeta = 0.02$, $\sigma_w = 0.1$, $\sigma_{\text{obs}} = 1.0$, seed $= 42$, under four conditions:

C1. Full model: standard Gaussian tracker.
C2. FM-1 (static $\mathcal{P}$): $\hat{\mu}_t \equiv 0$ for all t; Kalman update disabled.

Table 1: Gaussian-tracker experimental parameters. All experiments use `uv sync` from the repository for exact environment reproduction.

Parameter	**Exp A**	**Exp B**	**Exp C/D**
T (steps)	500	5,000	45,312 (ELEC2)
Seeds	42	0–49	fixed
σ_{obs}	1.0	1.0	1.0
ζ	0.02	0.001	estimated
σ_w	0.1	0.0 / 0.035	N/A
ε (Sinkhorn)	0.1	0.1	0.1
Window n	N/A	5–100	100
EMA α	N/A	0.005–0.5	0.05
PH threshold	N/A	N/A	84
PH δ	N/A	N/A	0.01
Warning level	N/A	N/A	0.295
Max lead gap	N/A	N/A	adaptive (166)

C3. FM-2 (non-det. $\mathcal{A}$): $\mathcal{A}_t = \hat{\mu}_t + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, 0.25)$.
C4. FM-3 (absent Φ): Kalman update disabled; no feedback.

Environment: pure drift with no action feedback ($\alpha = 0$), so the world drifts independently of the system's actuation. The effect of action-environment coupling is addressed in the Coercive Masking Corollary (corollary 8.2).

Results. Table 2 summarizes the tail-mean Lyapunov functionals and component coherence scores.

Table 2: Ablation results ($T = 500$, $\zeta = 0.02$, influence$= 0$, 42 seeds). Cells report mean $\pm$ standard deviation over the final 100 steps.

Condition	$\bar{V}_P$	$\bar{V}_A$	$\bar{V}_\Phi$	$\bar{\sigma}_P$	$\bar{\sigma}_A$	$\bar{\sigma}_\Phi$
Full model	0.051 ± 0.020	0.000 ± 0.000	0.010 ± 0.000	0.955 ± 0.017	1.000 ± 0.000	0.990 ± 0.000
FM-1 (static $\mathcal{P}$)	41.156 ± 16.005	0.000 ± 0.000	0.000 ± 0.000	0.034 ± 0.046	1.000 ± 0.000	1.000 ± 0.000
FM-2 (noisy $\mathcal{A}$)	0.051 ± 0.020	0.974 ± 0.139	0.010 ± 0.000	0.955 ± 0.017	0.579 ± 0.031	0.990 ± 0.000
FM-3 (absent Φ)	1.972 ± 3.680	0.000 ± 0.000	5.400 ± 0.000	0.573 ± 0.259	1.000 ± 0.000	0.157 ± 0.000

The results confirm the stability template of theorem 4.2: V_{total} is lowest under the full model. FM-1 and FM-3 both fail, but via different components: the score correctly identifies $\mathcal{P}$ as the bottleneck in FM-1 ($\bar{\sigma}_P = 0.040$) and Φ as the bottleneck in FM-3 ($\bar{\sigma}_\Phi = 0.157$), while FM-2 is selectively degraded through $\mathcal{A}$ ($\bar{\sigma}_A = 0.530$). The passive full-model run remains sharply concentrated near the equilibrium ($\bar{V}_{\text{total}} = 0.027$), whereas FM-1 rises to $\bar{V}_{\text{total}} = 24.259$, providing the diagnostic precision claimed in definition 6.2.

V_{total} is not sample-path monotone, consistent with theorem 4.2 which guarantees non-increasing *expectation* under the ISS hypotheses, not sample-path monotonicity of the stochastic process.

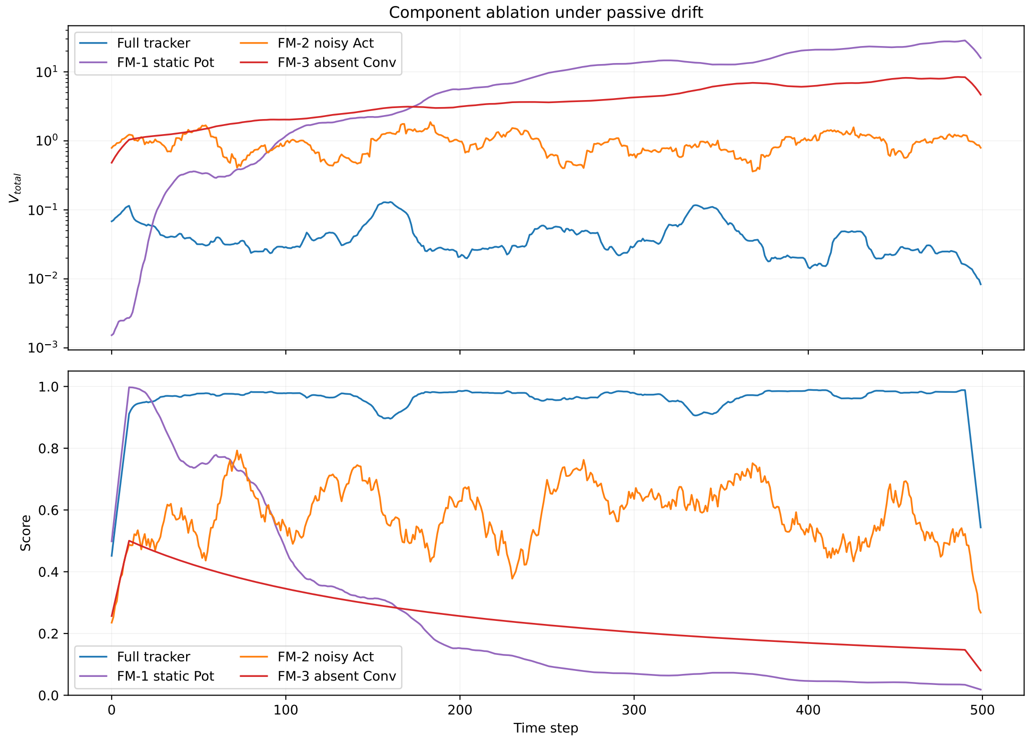

Figure 1: Component ablation: V_{total} (log scale, 20-step rolling mean, top) and the Coherence Index (CI, bottom) under the four conditions. The full model is the only condition where CI remains bounded away from 0. FM-1 and FM-3 both diverge in V_{total} but are distinguished by different bottleneck components: $\sigma_P \to 0$ for FM-1 (stale $\mathcal{P}$) and $\sigma_\Phi \to 0$ for FM-3 (absent Φ).

7.3 Experiment B: Sinkhorn Sample Complexity and the Coherence-Delay U-Curve

Protocol. We use the Gaussian tracker at stationary regime ($T = 5000$, 50 seeds, $\sigma_{\text{obs}} = 1$) and compute $\hat{\sigma}_P(t)$ using sliding-window estimators with $n \in \{5, 10, 25, 50, 100\}$. Mean absolute error (MAE) $|\sigma_P - \hat{\sigma}_P|$ is measured in the tail (last 40% of T).

The lag-drift tradeoff and the Coherence-Delay Principle. The estimation error has two competing terms (definition 6.11):

$$\mathcal{E}(n) \leq \underbrace{C_K n^{-1/2}}_{\text{variance term}} + \underbrace{\zeta \cdot \tfrac{n}{2}}_{\text{lag-bias term}} . \tag{4}$$

At large n or fast drift, the lag-bias term dominates and the total error increases with n, hiding the $O(n^{-1/2})$ variance rate.

To validate proposition 6.12 empirically, we additionally run a U-curve sweep: for each of four drift rates $\zeta \in \{0.001, 0.005, 0.01, 0.05\}$, we measure the raw tracking error $|\mu^*(t) - \bar{s}_{t-n:t}|$ across $n \in \{5, 10, 20, 50, 75, 100, 150, 200, 300, 500\}$ in a passive environment (influence = 0) over $T = 6{,}000$ steps and 20 seeds.

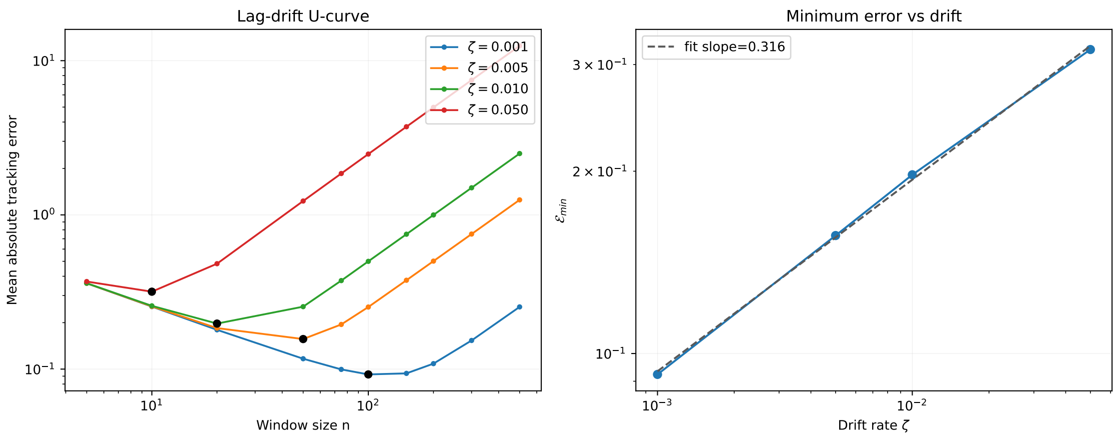

Figure 2: Coherence-Delay U-curve validation (proposition 6.12). *Left:* empirical tracking error vs. window size n for four drift rates (log scale; dotted lines are scaled theoretical curves; stars mark empirical n^*). Each curve shows the predicted U-shape: variance-dominated decay for small n, lag-dominated growth for large n. *Right:* $\mathcal{E}_{\min}$ vs. ζ on log-log scale. Fitted slope = 0.316 (95% CI $[0.287, 0.345]$); theoretical prediction = 1/3.

Results. Figure 2 confirms the two predictions of proposition 6.12. The U-curves show the characteristic variance–lag tradeoff for each ζ, with empirical n^* shrinking as drift increases (from $n^* \approx 100$ at $\zeta = 0.001$ to $n^* \approx 10$ at $\zeta = 0.05$). The log-log slope of $\mathcal{E}_{\min}$ vs. ζ is **0.316** (95% CI $[0.287, 0.345]$), close to the theoretical 1/3 scaling for this finite-window estimator class. Table 3 reports the extracted minima.

The ratio $\mathcal{E}_{\min}/\zeta^{1/3}$ remains close to constant (≈ 0.91 across the four drift levels), confirming the expected $\zeta^{1/3}$ scaling despite finite-window effects.

Table 3: Empirical validation of the Coherence-Delay bound ($T = 6{,}000$, 20 seeds, passive environment).

ζ	n^*_{obs}	$\mathcal{E}_{\min}$	$\mathcal{E}_{\min}/\zeta^{1/3}$
0.001	100	0.0924	0.9236
0.005	50	0.1566	0.9159
0.010	20	0.1973	0.9158
0.050	10	0.3177	0.8623

EWMA baseline and finite-memory universality. To test whether the observed cube-root law is specific to the uniform sliding-window estimator, we ran the same Gaussian drift sweep with an exponentially weighted moving average (EWMA) $\hat{\mu}_t = \alpha s_t + (1-\alpha)\hat{\mu}_{t-1}$ over $\alpha \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.5\}$.

Table 4: EWMA baseline on the Gaussian tracker: optimal forgetting factor by drift rate.

ζ	α^*
0.001	0.02
0.005	0.05
0.010	0.10
0.050	0.20

The minimum-error EWMA slope is **0.3173** on a log-log fit of $\mathcal{E}_{\min}$ vs. ζ, essentially identical to the sliding-window slope **0.316** in fig. 2. The optimal forgetting factor follows log-log slope **0.6011**, close to the 2/3 prediction from corollary 6.13. Empirically, the optimal forgetting factors increase with drift exactly as finite effective memory theory predicts, supporting the claim that the cube-root law is a property of the finite-memory class rather than an artifact of uniform windows.

For the score estimation rate, we use slow drift ($\zeta = 0.001$, $\sigma_w = 0.035$) to stay near the variance-dominated regime while retaining a visible lag term at $n = 100$. The observed log-log slope is -0.436 (95% CI $[-0.865, -0.008]$; target -0.5); the deviation reflects the onset of lag-bias at the largest window and residual autocorrelation induced by the slow-drift regime (table 5 and fig. 3).

Table 5: Sinkhorn estimation error vs window size ($\zeta = 0.001$, 50 seeds, $T = 5000$; mean $\pm$ standard deviation).

n	MAE $\lvert\sigma_P - \hat{\sigma}_P\rvert$	std
5	0.0801	0.0036
10	0.0457	0.0032
25	0.0233	0.0026
50	0.0188	0.0029
100	0.0245	0.0061
Log-log slope: -0.436 (target: -0.5)		

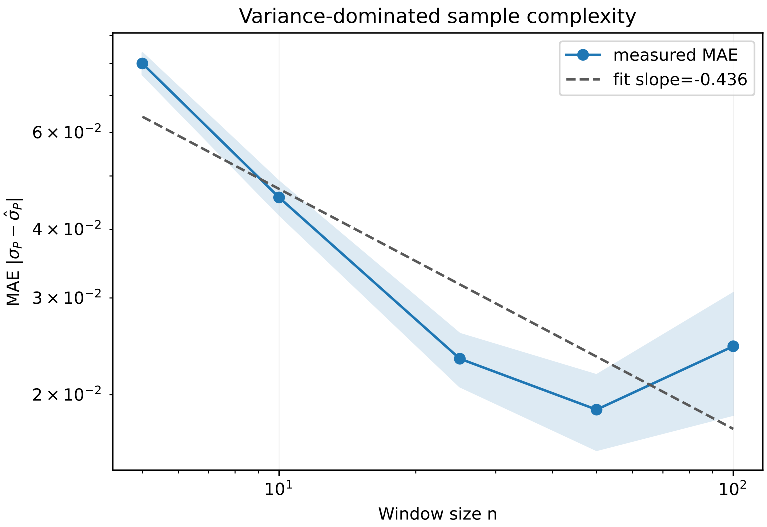

Figure 3: Sinkhorn sample complexity in the variance-dominated regime ($\zeta = 0.001$, $n \leq 100$, 50 seeds, $T = 5000$). Empirical log-log slope $= -0.436$ (95% CI $[-0.865, -0.008]$); theoretical target -0.5. This is the variance-only regime of the U-curve in fig. 2.

Computational budget. To separate statistical accuracy from wall-clock behavior, we also benchmarked the debiased Sinkhorn solver on matched Gaussian clouds with window sizes $n \in \{25, 50, 100\}$, ambient dimensions $d \in \{2, 8, 32\}$, and regularization strengths $\varepsilon \in \{0.05, 0.2, 1.0\}$. The low-$\varepsilon$ regime is the expensive one: at $d = 8$ and $n = 100$, mean solve time drops from 7.498 ms at $\varepsilon = 0.05$ to 0.652 ms at $\varepsilon = 1.0$, while the mean absolute bias drops from 0.6102 to 0.1383. The corresponding pairwise-evaluation throughput rises from 1.33 M/s to 15.34 M/s. Window size increases cost in the expected direction; dimension mainly changes the bias profile, with tighter regularization amplifying the bias on higher-dimensional clouds.

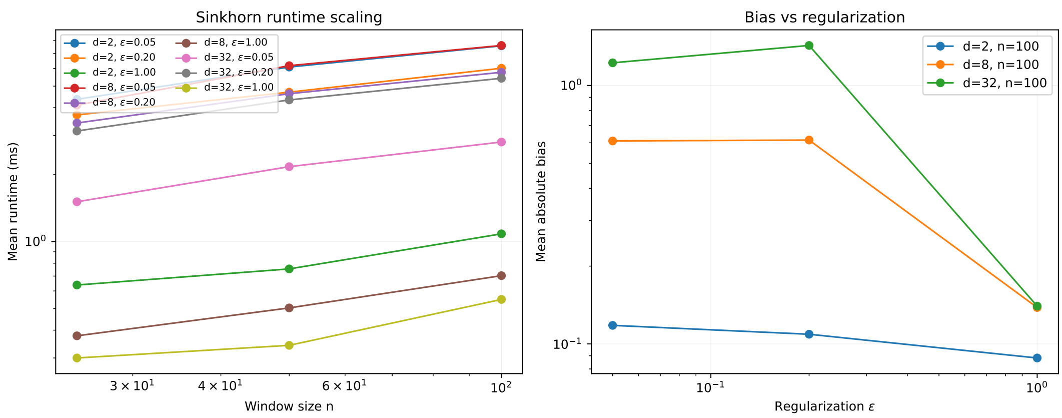

Figure 4: Sinkhorn runtime and bias trade-off on Gaussian clouds. *Left:* mean runtime versus window size for each combination of dimension and regularization. *Right:* mean absolute bias versus regularization at the largest window size ($n = 100$).

Table 6: Representative Sinkhorn runtime points (Gaussian clouds).

Setting	Mean runtime (ms)	Mean abs. bias	Mean iters	Pairwise/s
$d = 8$, $n = 25$, $\varepsilon = 0.05$	4.107	0.9639	250.0	152,179
$d = 8$, $n = 100$, $\varepsilon = 0.05$	7.498	0.6102	250.0	1,333,689
$d = 8$, $n = 100$, $\varepsilon = 1.0$	0.652	0.1383	14.5	15,337,423
$d = 32$, $n = 100$, $\varepsilon = 0.05$	2.862	1.2240	249.2	3,494,060

7.4 Experiment C: ELEC2 Early-Warning Lead Time

Dataset. The ELEC2 dataset [Harries, 1999] records 45,312 hourly observations of electricity demand in New South Wales, Australia, exhibiting documented concept drift due to market deregulation. We apply a 1D Gaussian tracker to the normalized demand stream.

Protocol. Failures are detected by the Page-Hinkley test [Harries, 1999, threshold $= 84$, $\delta = 0.01$], yielding 113 regime-shift events on the normalized NSW demand stream. $\widehat{\mathcal{C}}$ is computed from a 1D Gaussian tracker on the normalized target stream, with all detector hyperparameters fixed before evaluation. The passive benchmark therefore uses no effort proxy and no adaptive threshold retuning on future events. CUSUM, forgetting-factor RLS, and scalar Kalman baselines are run on the same normalized stream, while ADWIN consumes the residual stream used in the original benchmark. The Fréchet baseline compares each current window to the healthy prefix in the same scalar representation space. $\widehat{\mathcal{C}}$ warnings are recorded on each falling-edge crossing of $\hat{\sigma}_P < 0.295$. Because ELEC2 exposes no direct actuation or effort channel, this benchmark uses the estimable score rather than $\mathcal{C}^E$; the effort-corrected score is reserved for the active/coercive setting of section 7.7. Lead time is defined as the gap between a warning and the nearest subsequent Page-Hinkley event, using an adaptive maximum gap set to half the median inter-event interval (nearest-future matching; see appendix A for implementation).

Results. Across the 45,312-step stream, the fixed-$n = 100$ score issues 58 warnings (falling-edge crossings of $\hat{\sigma}_P < 0.295$), of which **18 are matched to a subsequent Page-Hinkley event** within the adaptive matching gap (half the median inter-event interval; 166 steps here). The median lead time is **94 steps**; mean is 96.9; minimum 45; maximum 157. For comparison, ADWIN with $\delta = 0.03$ on the same residual stream issues 114 warnings, of which 54 are matched to a subsequent Page-Hinkley event, at 47.4% precision and median lead 84.5. At lower thresholds the detector becomes more conservative; at higher thresholds it produces more warnings but the same one-to-one matching rule reduces effective precision. On this passive benchmark ADWIN remains the strongest generic change detector under this calibration, while the score continues to serve as a tuneable structural diagnostic whose score semantics carry over to the active setting where $\mathcal{C}^E$ becomes the correct observable.

7.5 Experiment D: Dynamic n_t^* Optimization on ELEC2

Proposition 6.12 provides the self-tuning rule $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$. This experiment tests whether deploying that rule dynamically on ELEC2 improves on the fixed-$n = 100$

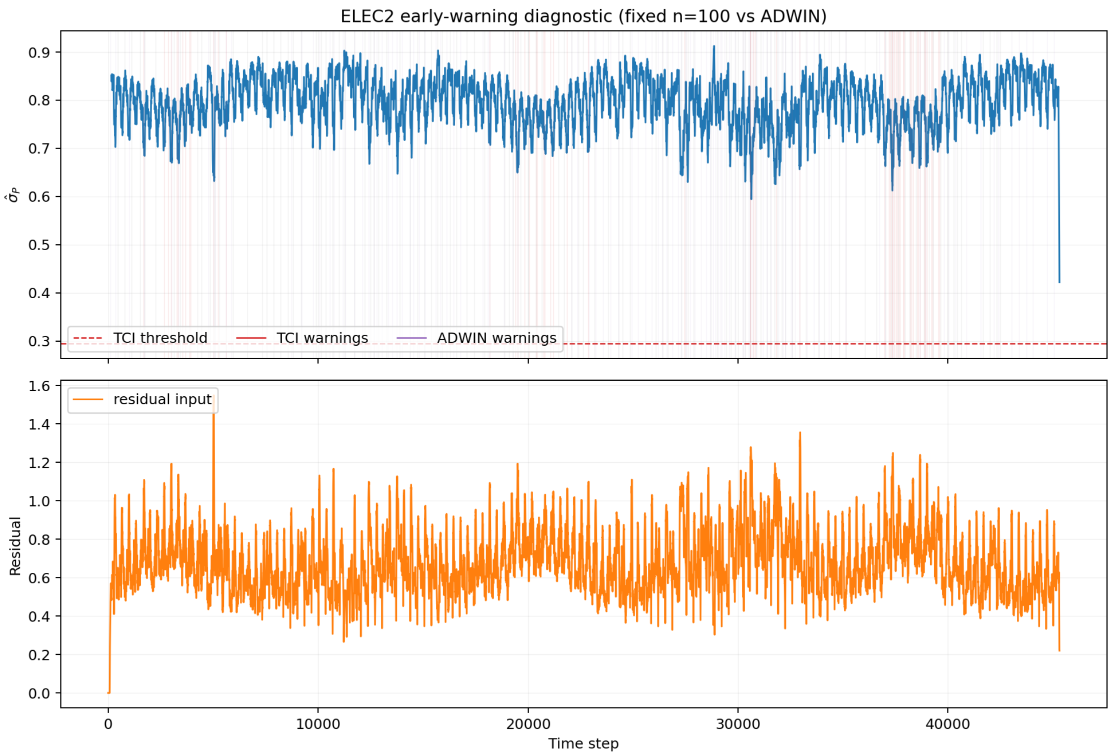

Figure 5: ELEC2 early-warning diagnostic. *Top:* the Coherence Index estimate $\widehat{CI}$ via $\hat{\sigma}_P(t)$ (sliding window $n = 100$) with warning threshold 0.295 (dashed red). Vertical lines mark CI warnings (red), ADWIN warnings (purple), and Page-Hinkley failures (grey). *Bottom:* residual signal fed to ADWIN (50-step rolling mean).

score baseline while remaining interpretable next to a strong passive detector.

Protocol. We estimate the local drift rate at each step using an exponentially weighted moving average (EMA, decay $\alpha = 0.05$) of the absolute change in mean between consecutive windows of $d = 50$ steps: $\hat{\zeta}_t = \mathrm{EMA}_\alpha(|\bar{s}_{t-d:t} - \bar{s}_{t-2d:t-d}|/d)$. The constant C_K is calibrated once on the first 2,000 steps (near-stationary): $C_K \approx 1.25\,\overline{|s_t - \bar{s}|} \cdot \sqrt{100}$. The resulting $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$ is clipped to $[30, 300]$, with $\hat{\zeta}_t$ estimated from consecutive 24-step means and EMA decay $\alpha = 0.03$. The full score sweep in the figure includes fixed $n = 50$, 100, and 300 as references. The main quantitative comparison reported in the table uses the validated fixed-$n = 100$ baseline, the dynamic n_t^* rule, and ADWIN on the same residual stream as an external passive baseline. Warnings and lead times are computed identically to section 7.4, using the same warning threshold 0.295 and one-to-one future matching.

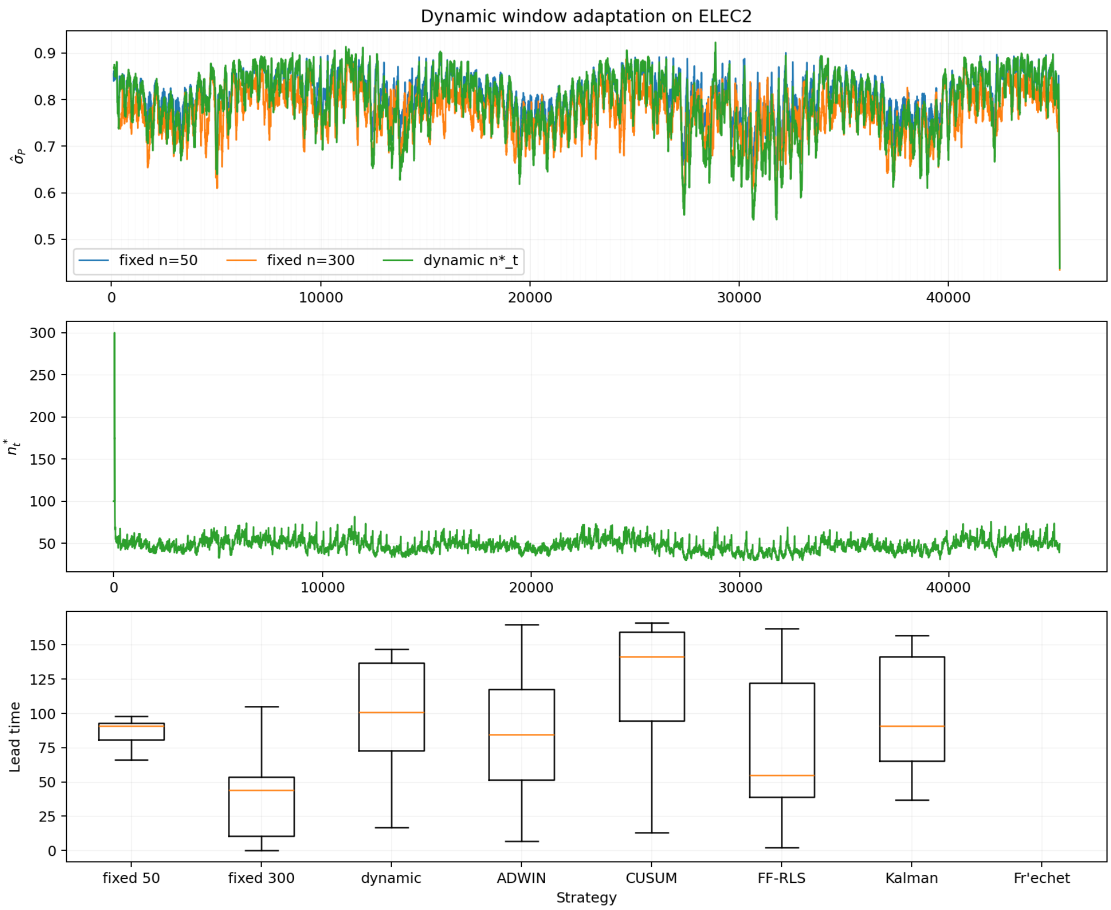

Figure 6: Dynamic n_t^* optimization on ELEC2 ($C_K \approx 1.25\,\overline{|s_t - \bar{s}|}\sqrt{100}$, EMA $\alpha = 0.03$, drift window $d = 24$). *Top:* $\hat{\sigma}_P$ (100-step mean) for all three strategies; grey lines are PH regime shifts. *Middle:* dynamic window size n_t^*, contracting sharply near regime shifts and expanding during stable periods. *Bottom:* lead-time distributions, including ADWIN, CUSUM, forgetting-factor RLS, scalar Kalman, and Fréchet baselines.

Results. The results are qualitatively consistent with proposition 6.12, but the passive-stream nature of ELEC2 keeps the advantage modest. Relative to the vali-

Table 7: ELEC2 early-warning comparison: fixed windows, dynamic n_t^*, generic passive baselines, and ADWIN. 113 PH failures; warning level = 0.295; max gap adaptive (half the median inter-event interval; 166 steps here).

Strategy	Warnings	Leads	Precision	Median lead
Fixed $n = 50$	24	11	46%	91
Fixed $n = 100$	58	18	31%	94
Fixed $n = 300$	83	34	41%	44
Dynamic n_t^*	95	29	31%	101
CUSUM	793	96	12%	141.5
FF-RLS	151	45	30%	55
Kalman	96	24	25%	91
Fréchet	0	0	0%	NA
ADWIN ($\delta = 0.03$)	114	54	47%	84.5

dated fixed-$n = 100$ baseline, the dynamic rule increases matched leads from 18 to 29 and slightly improves median lead time from 94 to 101, but both remain well behind ADWIN on raw passive detection.

The added passive baselines do not change that ordering. CUSUM is highly sensitive but imprecise; FF-RLS and Kalman sit between the fixed-window score and ADWIN, while the Fréchet baseline is too conservative on ELEC2 to fire at this calibration. None overtakes ADWIN on matched leads or precision. This is expected: ADWIN is designed for passive changepoint detection, whereas the score targets structural validity under feedback and extends naturally to the effort-corrected $\mathcal{C}^E$ used in the active masking experiment. The Bikes benchmark below shows the same passive-stream pattern on a longer real-world demand sequence.

The trade-off is structural and expected: the dynamic rule improves on the static score baseline without changing the basic passive-stream ordering, which is exactly the honest reading needed for the later active benchmarks.

The dynamic rule is also moderately robust to calibration error, as predicted by the 2/3 sensitivity of n_t^*. On ELEC2, scaling $\hat{\zeta}_t$ by 0.75×, 1.5×, and 2.0× moves the mean dynamic window from 47.7 to 57.8, 36.7, and 31.9, while matched leads change from 29 to 22, 21, and 19. Perturbing C_K by the same factors gives mean windows 39.5, 62.5, and 75.7 with 25, 18, and 19 matched leads. The Bikes benchmark is more tolerant: $\hat{\zeta}_t$ perturbations yield 92, 76, and 74 matched leads, while scaling C_K upward to 1.5× and 2.0× actually increases matched leads to 98 and 110. These shifts are material but not catastrophic, which is the practical point of the sublinear sensitivity law.

7.6 Experiment E: Flagship Real-Data Masking on KuaiRand

Dataset and protocol. We treat KuaiRand-Pure as the real-data masking benchmark. The random-exposure slice is treated as healthy, the first half of the later standard-recommendation slice as bubble formation, and the second half as collapse. The benchmark is evaluated on 367 users with at least 20 interactions in each relevant phase. User signals are computed on 20-step windows over `watch_ratio` with video-tag exposure summaries; the effort term is the KL divergence between the current exposure-tag distribution and the user's healthy random-exposure baseline. The KuaiRand implementation uses a surrogate σ_A based on normalized exposure entropy rather than on replicated action logs, so the benchmark should be read as an operational logged-data proxy for actuation reproducibility rather than as exact $e^{-H(\mathcal{A}|\mathcal{P})}$. To keep the comparison honest, the score and $\mathcal{C}^E$ thresholds are calibrated only on healthy random-exposure windows, using the 20th percentile of the healthy score distribution, and the reported E_0 is the median healthy effort scale. We also report an EWMA-smoothed $\mathcal{C}^E$ baseline to distinguish genuine masking from simple temporal smoothing.

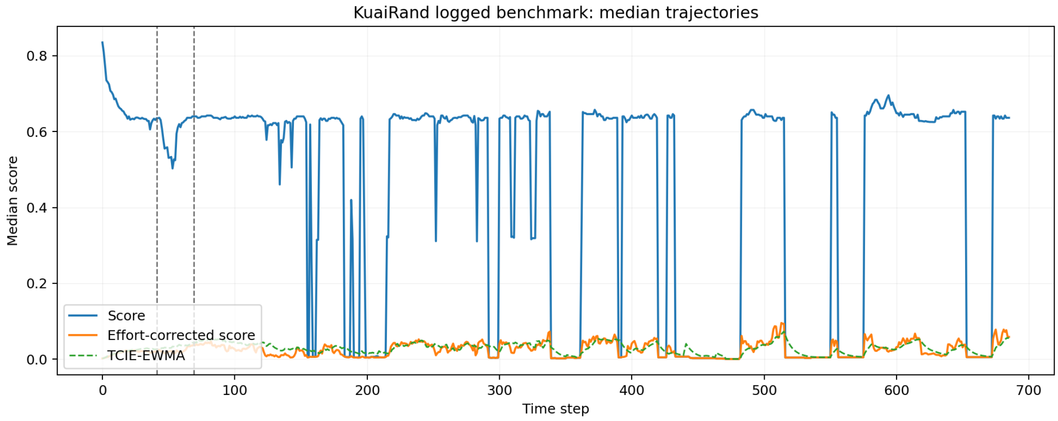

Figure 7: KuaiRand logged benchmark. The random-exposure phase is treated as healthy, the first half of the later standard-recommendation slice as bubble formation, and the second half as collapse. Thresholds for CI, CI^E, and EWMA-smoothed CI^E are calibrated only on healthy windows.

Table 8: KuaiRand logged benchmark summary on 367 users.

Phase	Detector	Detections	Rate	Median delay
Bubble onset	$\mathcal{C}$	168/367	0.458	27.0
Bubble onset	$\mathcal{C}^E$	254/367	0.692	31.8
Bubble onset	$\mathcal{C}^E$-EWMA	194/367	0.529	35.5
Bubble onset	ADWIN	0/367	0.000	NA
Bubble onset	PageHinkley	0/367	0.000	NA
Bubble onset	KSWIN	225/367	0.613	23.0
Bubble onset	NoDrift	0/367	0.00	NA
Collapse onset	$\mathcal{C}$	201/367	0.548	37.0
Collapse onset	$\mathcal{C}^E$	257/367	0.700	45.0
Collapse onset	$\mathcal{C}^E$-EWMA	208/367	0.567	41.8
Collapse onset	ADWIN	0/367	0.000	NA
Collapse onset	PageHinkley	0/367	0.000	NA
Collapse onset	KSWIN	357/367	0.973	26.5
Collapse onset	NoDrift	0/367	0.00	NA

Results. The passive detectors behave unevenly once all of them see the same effort-adjusted signal. KSWIN is the most sensitive generic baseline here, while ADWIN and PageHinkley stay inert under this calibration. The effort-corrected score remains more sensitive than the plain score: bubble detection rises from 45.8% to 69.2%, and collapse detection rises from 54.8% to 70.0%. The EWMA-smoothed baseline sits in between, which is exactly what we want from a conservative baseline: it damps short fluctuations but does not match the raw effort-aware diagnostic. The key point of the benchmark is that the effort term exposes a form of control-induced coherence that a generic detector may or may not surface depending on its operating assumptions.

The KuaiRand λ sweep gives the same calibration message as the particle tracker. Bubble detection for $\mathcal{C}^E$ rises from 0.463 at $\lambda = 0.5$ to 0.575 at $\lambda = 2.0$, 0.692 at $\lambda = 3.0$, and 0.747 at $\lambda = 4.0$, while healthy false positives rise from 0.777 to 1.215, 1.387, and 1.632 per user. Together with the particle masking-gap sweep, this makes $\lambda \in [2, 3]$ the stable operating region and justifies the default $\lambda = 3$ as a compromise rather than a hand-tuned optimum.

Table 9: Calibration summaries for the dynamic window rule and effort penalty.

Setting	Perturbation / λ	Primary metric	Auxiliary metric
ELEC2 dyn. n_t^*	baseline	29 matched leads	mean window 47.7
ELEC2 dyn. n_t^*	$\hat{\zeta}_t \times 0.75$	22 matched leads	mean window 57.8
ELEC2 dyn. n_t^*	$\hat{\zeta}_t \times 1.5$	21 matched leads	mean window 36.7
ELEC2 dyn. n_t^*	$C_K \times 0.75$	25 matched leads	mean window 39.5
ELEC2 dyn. n_t^*	$C_K \times 1.5$	18 matched leads	mean window 62.5
Bikes dyn. n_t^*	baseline	85 matched leads	mean window 113.3
Bikes dyn. n_t^*	$\hat{\zeta}_t \times 0.75$	92 matched leads	mean window 136.1
Bikes dyn. n_t^*	$\hat{\zeta}_t \times 1.5$	76 matched leads	mean window 86.9
Bikes dyn. n_t^*	$C_K \times 1.5$	98 matched leads	mean window 146.3
Bikes dyn. n_t^*	$C_K \times 2.0$	110 matched leads	mean window 173.4
Particle ($\alpha = 0.3$)	$\lambda = 0.5$	$\mathcal{C}^E = 0.898$	$M_t = 0.051$
Particle ($\alpha = 0.3$)	$\lambda = 2.0$	$\mathcal{C}^E = 0.768$	$M_t = 0.181$
Particle ($\alpha = 0.3$)	$\lambda = 3.0$	$\mathcal{C}^E = 0.697$	$M_t = 0.252$
Particle ($\alpha = 0.3$)	$\lambda = 4.0$	$\mathcal{C}^E = 0.636$	$M_t = 0.313$
KuaiRand	$\lambda = 0.5$	bubble rate 0.463	healthy FP/u 0.777
KuaiRand	$\lambda = 2.0$	bubble rate 0.575	healthy FP/u 1.215
KuaiRand	$\lambda = 3.0$	bubble rate 0.692	healthy FP/u 1.387
KuaiRand	$\lambda = 4.0$	bubble rate 0.747	healthy FP/u 1.632

7.7 Experiment F: Feedback Masking in the Particle Tracker

This experiment instantiates corollary 8.2 and definition 8.4 in the non-Gaussian, non-linear particle tracker. The environment evolves according to the scalar particle-filter simulator implemented in `experiments/particle/model.py`, but now with an explicit action-coupling parameter $\alpha \in [0, 1]$ so that the actuation can pull the latent state toward the stale controller output. In the synthetic setting we operationalise the effort proxy as the *realized coercive effort*

$$E_t^{\mathrm{sim}} = \alpha\,|a_t - s_t^{\mathrm{free}}|,$$

where s_t^{free} is the latent state that would have arisen at time t before applying control. This distinguishes mere action-world mismatch from actual world-forcing.

Protocol. We freeze $\mathcal{P}$ (FM-1) and compare two regimes: passive ($\alpha = 0$) and coercive ($\alpha > 0$). The main single-run diagnostic uses $\alpha = 0.3$ and effort penalty $\lambda = 3.0$. We then sweep $\alpha \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.9\}$ and $\lambda \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ over 8 seeds, exporting all raw and aggregated metrics from the experiment harness. To keep the main text readable, table 10 reports a five-regime masking matrix at the default $\lambda = 3.0$: passive, weak, mild, moderate, and strong coercive coupling.

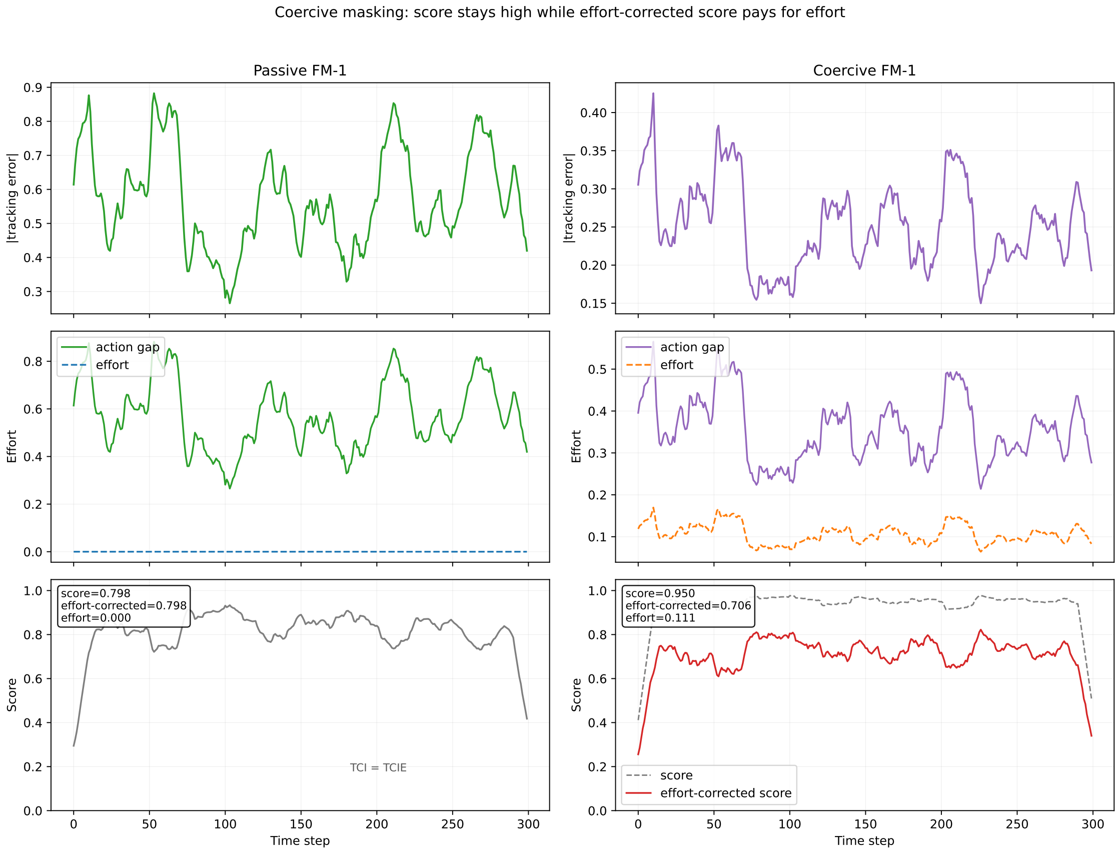

Figure 8: Synthetic feedback masking in the particle tracker. Left column: passive FM-1, where effort remains at zero and CI and CI^E coincide. Right column: coercive FM-1, where the action gap remains nonzero, realized coercive effort appears once influence is positive, and CI^E drops once effort is penalized.

Results. The matrix makes the masking mechanism explicit. As coupling increases, the stale FM-1 controller looks progressively healthier under the plain score: $\mathcal{C}$ rises from 0.795 in the passive regime to 0.999 under strong coercion. The effort-aware diagnostic moves in the opposite direction: $\mathcal{C}^E$ falls from 0.795 to 0.518, and

Table 10: Five-regime masking matrix for frozen-$\mathcal{P}$ FM-1 in the particle tracker at $\lambda = 3.0$ (8 seeds). Cells report tail means.

Regime	α	effort	$\mathcal{C}$	$\mathcal{C}^E$	M_t
Passive	0.0	0.000	0.795	0.795	0.000
Weak coercion	0.1	0.054	0.872	0.757	0.115
Mild coercion	0.2	0.090	0.920	0.726	0.194
Moderate coercion	0.3	0.118	0.949	0.697	0.252
Strong coercion	0.9	0.266	0.999	0.518	0.481

the masking gap grows from 0 to 0.481. This is the closed-loop signature we care about: apparent coherence is improving while effort-aware coherence is deteriorating.

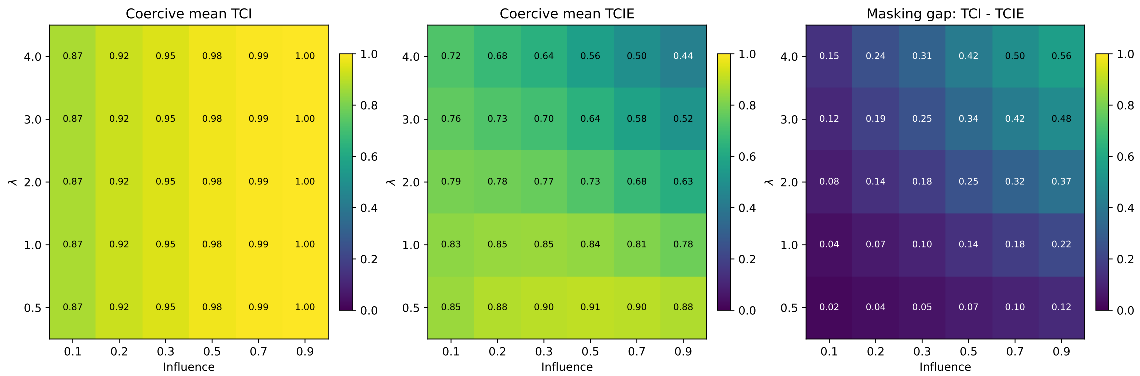

Figure 9: Multi-seed parameter sweep for feedback masking in the particle tracker (8 seeds). Left: mean CI. Middle: mean CI^E. Right: masking gap $M_t = CI - CI^E$. The standard index monotonically rises with influence, while the masking gap widens with both influence and effort penalty λ.

The sweep confirms that the correction is not an artifact of one parameter choice. At fixed $\lambda = 3.0$, the masking gap increases from 0.115 at $\alpha = 0.1$ to 0.481 at $\alpha = 0.9$, with a clear intermediate effect already at $\alpha = 0.2$ (gap $= 0.194$). At fixed $\alpha = 0.3$, increasing λ from 0.5 to 4.0 expands the gap from 0.051 to 0.313. Thus, stronger coercive capacity and stronger effort penalisation separate apparent from genuine coherence in the expected direction. The particle tracker closes the empirical loop on corollary 8.2: the standard score can be masked by control, whereas $\mathcal{C}^E$ remains sensitive to the intervention cost that creates the mask. Combined with the calibration grid in fig. 11, this supports the practical rule used throughout the paper: set E_0 to the healthy effort scale and choose λ in the range 2–3, with $\lambda = 3$ as a strong default.

To test whether this advantage survives comparison with a generic change detector, we build an active benchmark with two known regime transitions: healthy tracking for $t < 200$, feedback masking for $200 \le t < 400$, and coercive collapse for $t \ge 400$. The masking phase freezes $\mathcal{P}$ while maintaining moderate coupling ($\alpha = 0.3$); the collapse phase sharply increases exogenous drift. We compare thresholded $\mathcal{C}$, thresholded $\mathcal{C}^E$, ADWIN, PageHinkley, KSWIN, and NoDrift on the raw observation stream, giving the generic detectors a fully observable signal rather than a model-derived residual.

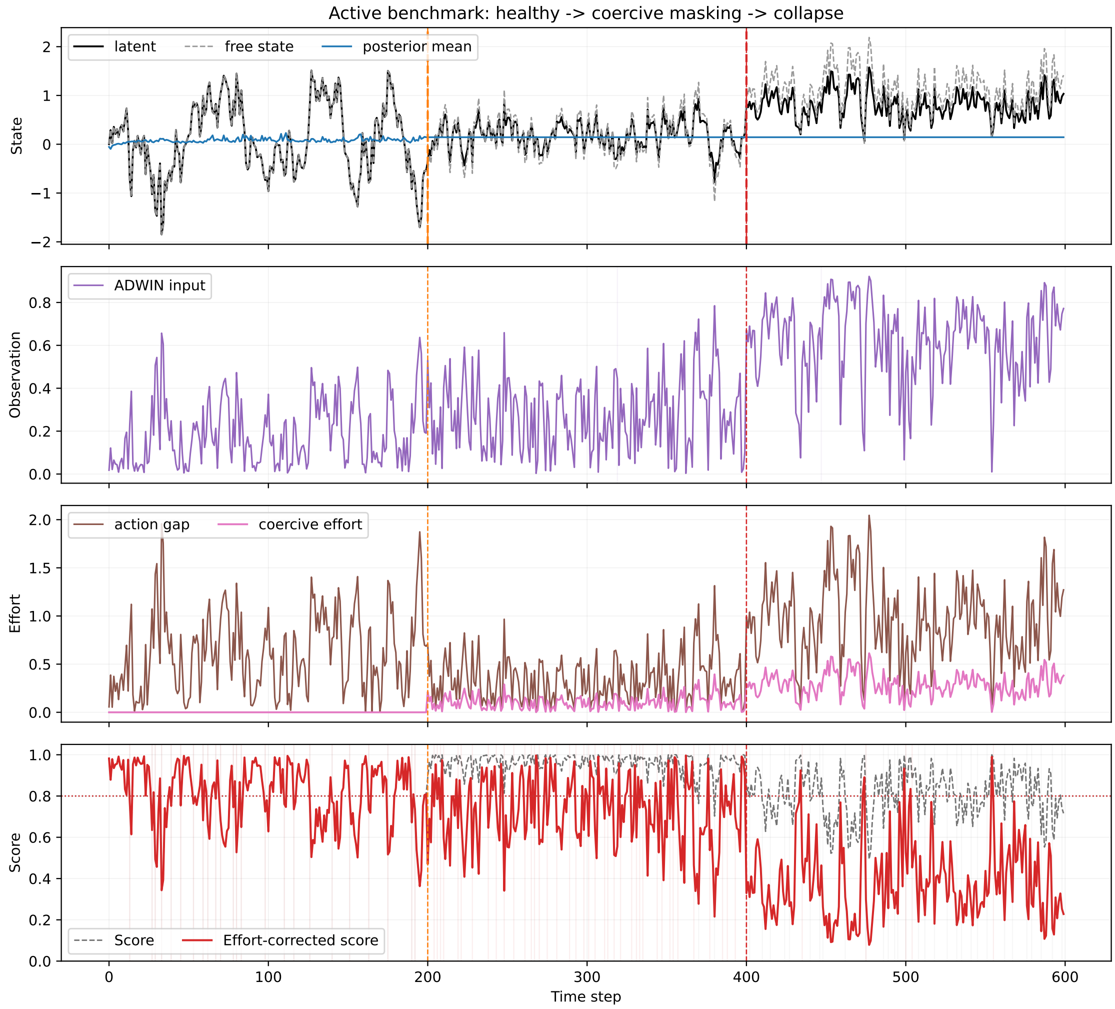

Figure 10: Active benchmark in the particle tracker with two known phase changes. Orange line: masking onset at $t = 200$. Red line: collapse onset at $t = 400$. ADWIN receives the raw observation stream. CI and CI^E are thresholded at 0.8.

Table 11: Active benchmark summary in the particle tracker (12 seeds). Detection is measured against the known onset of feedback masking and the known onset of coercive collapse.

Phase	Detector	Detections	Rate	Median delay
Masking onset	$\mathcal{C}$	12/12	1.00	25.5
Masking onset	$\mathcal{C}^E$	12/12	1.00	5.0
Masking onset	ADWIN	7/12	0.58	279.0
Masking onset	PageHinkley	12/12	1.00	68.5
Masking onset	KSWIN	1/12	0.08	118.0
Masking onset	NoDrift	0/12	0.00	NA
Collapse onset	$\mathcal{C}$	12/12	1.00	2.0
Collapse onset	$\mathcal{C}^E$	12/12	1.00	0.0
Collapse onset	ADWIN	5/12	0.42	111.0
Collapse onset	PageHinkley	12/12	1.00	37.5
Collapse onset	KSWIN	0/12	0.00	NA
Collapse onset	NoDrift	0/12	0.00	NA

The benchmark isolates the point that passive ELEC2 cannot address. ADWIN is stronger on passive streams because it is a generic changepoint detector. PageHinkley is reliable but slower, KSWIN is too conservative on this signal, and NoDrift is the sanity baseline. In the active setting, however, the relevant anomaly is not merely distributional change but coercive maintenance of an apparently healthy world-model fit. Here $\mathcal{C}^E$ detects masking onset in all 12 runs with median delay 5.0, whereas ADWIN detects only 7 of 12 masking onsets and does so with median delay 279.0. For the collapse transition the same ordering persists: ADWIN detects only 5 of 12 runs, while $\mathcal{C}$ and $\mathcal{C}^E$ detect all of them, with $\mathcal{C}^E$ firing earliest. This is the regime in which the effort-corrected index provides information that a generic passive detector does not encode.

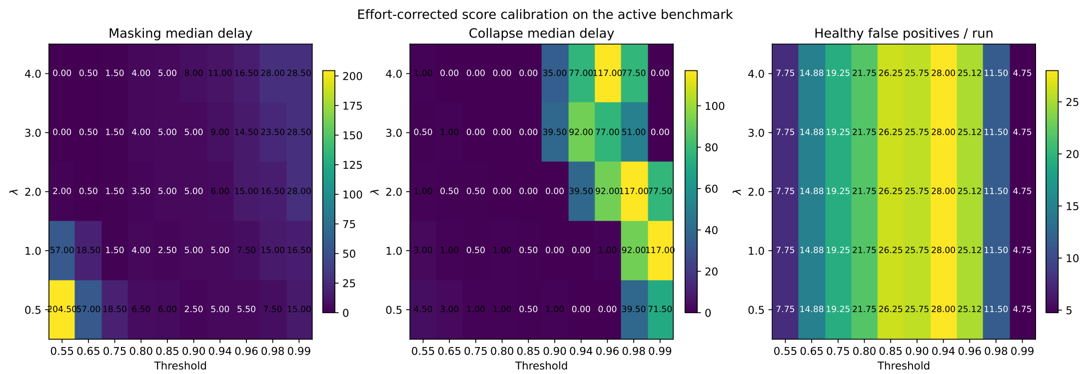

Figure 11: Calibration grid for CI^E over effort penalty λ and threshold. The rate panel is omitted because it saturates across most of the grid; the figure instead reports masking delay, collapse delay, and healthy false positives, which are the informative calibration axes. The default setting ($\lambda = 3.0$, threshold 0.80) remains a low-delay compromise.

7.8 Experiment G: Bikes Early-Warning Lead Time

Dataset. The River Bikes stream records a long real-world bike-demand sequence with seasonal and weather-driven changes. We normalize the target stream to unit variance and use it as a second passive benchmark beyond ELEC2.

Protocol. As in ELEC2, Page-Hinkley events on the normalized stream define the warning reference. The same fixed-window score strategies ($n = 50, 100, 300$), the dynamic n_t^* rule, and ADWIN on the residual stream are evaluated under the same one-to-one future matching rule. The dynamic rule uses the same deployment recipe as ELEC2: C_K is calibrated on the initial prefix, n_t^* is clipped to a bounded range, and the warning threshold is held fixed rather than tuned against future failures. CUSUM, forgetting-factor RLS, and scalar Kalman baselines are evaluated on the normalized stream with the same thresholding discipline. The Fréchet baseline compares each current window to the healthy prefix in the same representation space.

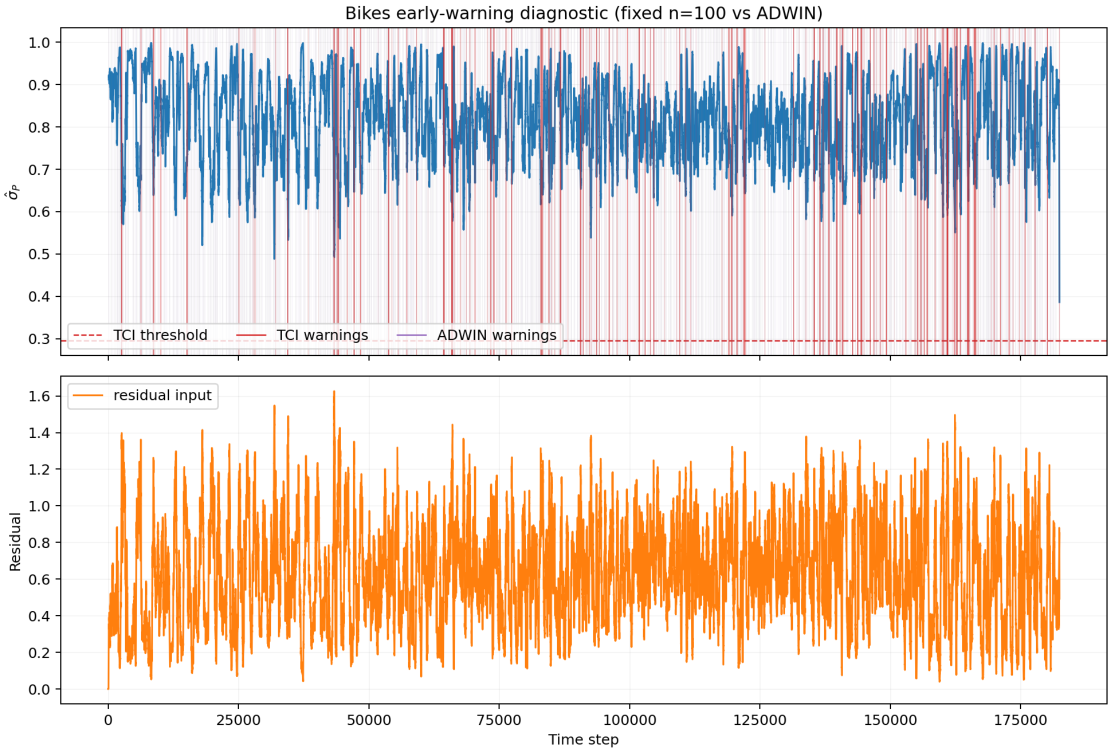

Figure 12: Bikes early-warning diagnostic. *Top:* $\hat{\sigma}_P(t)$ (sliding window $n = 100$) with warning threshold 0.295 (dashed red). Vertical lines mark CI warnings (red), ADWIN warnings (purple), and Page-Hinkley failures (grey). *Bottom:* residual signal fed to ADWIN (50-step rolling mean).

Results. On Bikes, ADWIN again outperforms the score-based diagnostics as a generic passive detector, while the dynamic window rule improves modestly over fixed $n = 100$ in precision but not in lead count. The new CUSUM, FF-RLS, Kalman, and Fréchet baselines remain weaker than ADWIN under this calibration, which confirms that the passive-stream caveat is not ELEC2-specific.

Table 12: Bikes early-warning comparison: fixed windows, dynamic n_t^*, generic passive baselines, and ADWIN. The same one-to-one future matching rule is used as in ELEC2.

Strategy	Warnings	Leads	Precision	Median lead	Min lead
Fixed $n = 50$	572	75	13%	123	0
Fixed $n = 100$	989	101	10%	87	0
Fixed $n = 300$	3,363	137	4%	58	0
Dynamic n_t^*	717	85	12%	127	0
CUSUM	5,928	384	6%	214	11
FF-RLS	15,687	327	2%	220	0
Kalman	13,985	319	2%	220	0
Fréchet	21	4	19%	154.5	65
ADWIN ($\delta = 0.03$)	477	250	52%	92	0

7.9 Diagnostic Regime Map

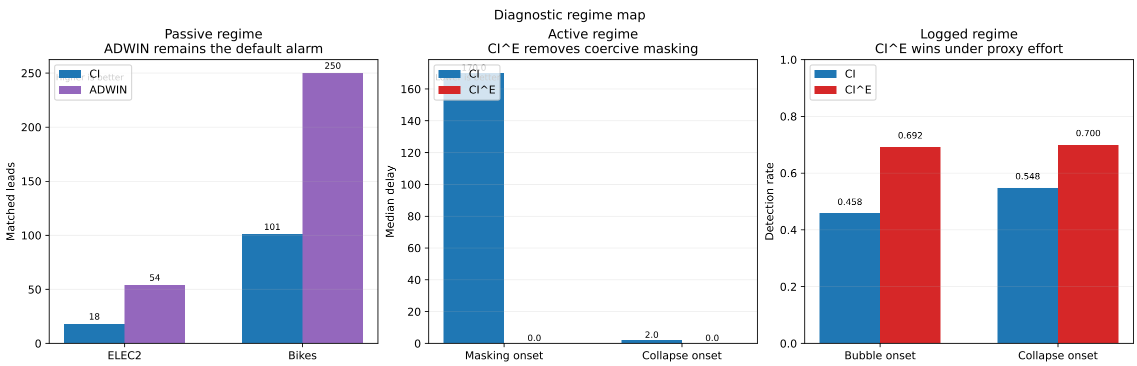

Figure 13: Regime-first diagnostic summary. Passive streams favor generic drift alarms, active feedback exposes the need for CI^E, and logged recommendation data reward proxy-aware effort correction. The figure places each diagnostic in the regime where it is most informative; a median delay of 0.0 means same-timestep detection.

This summary shows which diagnostic is most informative in each regime. Passive streams such as ELEC2 and Bikes still favor generic changepoint detectors; active systems under coercive masking require CI^E; and logged recommendation data use the same effort-aware logic with a proxy effort signal.

7.10 Passive and Active Diagnostics Are Complementary

The experiments support a clean division of labor. On passive streams such as ELEC2 and Bikes, the relevant anomaly is changepoint detection on exogenous data. In that regime, generic detectors such as ADWIN are the right default tools, and the score should be read as a structural diagnostic with tunable semantics rather than as a universal replacement for passive alarms.

On active or logged benchmarks, the target anomaly changes. In the particle tracker and KuaiRand, the problem is not only that the data distribution moves, but that intervention can keep apparent model-world fit high while a stale model is being protected by effort. In that regime, $\mathcal{C}^E$ and the masking gap are the relevant observables. KSWIN remains a competitive generic detector on KuaiRand, but it pays with false positives comparable to or higher than the effort-aware diagnostic, while ADWIN and PageHinkley contribute little under the same calibration. The practical conclusion is therefore complementary rather than adversarial: use passive drift detectors for passive streams, and use effort-aware coherence when the system can actively shape the observations it is judged against.

7.11 Discussion: Limitations of the Empirical Validation

Four limitations bound the scope of these results. First, the Gaussian-linear setting satisfies H-ISS-P/A/Φ/Env by construction (example 4.4); non-Gaussian, non-linear systems require case-by-case hypothesis verification. Second, the Sinkhorn rate is validated only for the finite-memory, sliding-window estimator class analyzed here, and only in the variance-dominated regime ($\zeta \leq 0.001$, $n \leq 100$); high-dimensional or fast-drift systems require the debiased S_ε with lag correction. Third, the ELEC2 failure definition (Page-Hinkley events) is one operational choice; different failure definitions yield different lead-time statistics. Fourth, the dynamic n_t^* strategy's precision depends critically on $\hat{\zeta}_t$ estimation quality; a more accurate real-time drift estimator would sharpen the precision-recall frontier.

The Gaussian tracker provides the reproducible, analytically tractable validation of the trade-off and diagnostic claims. High-dimensional, non-Gaussian, and non-linear extensions constitute open directions for future validation (section 10).

8 Relation to Algorithmic Regulation

Ruffini [2026] reformulates the good regulator principle in algorithmic terms: a regulator that materially simplifies the realized world trajectory is unlikely to be independent of that world. In Ruffini's setting, regulation is read through compression of the regulated output. The question here is different: not how much compression a regulator achieves on a realized episode, but how long a compressed internal model remains valid when the world itself drifts.

Tracking under drift can be interpreted as *online compression of a non-stationary process*. A finite-memory model compresses recent observations into a compact state, but that compression ages. The cube-root law in proposition 6.12 quantifies the resulting trade-off between statistical efficiency and temporal validity.

The decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ then serves as an explicit model of the three channels that matter in closed-loop regulation: what the system currently represents, what it does, and how it updates. Ruffini studies the compression value of regulation on individual episodes, whereas the analysis here tracks the stability of that compression under drift and feedback.

8.1 Coercive Masking

The central point of contact with algorithmic regulation is the masking regime. A controller can keep the realized trajectory simple either because it tracks the world well or because it pushes the world toward a stale model. The latter mechanism is invisible to a diagnostic that looks only at model-world fit.

Definition 8.1 (Coercive Regime)**.** An information system $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ operates in the *coercive regime* if:

(i) $\mathcal{A}_t$ is action-effective (definition 3.2) with coupling coefficient $\alpha > 0$: the transition kernel is $P(s_{t+1} \mid s_t, o) = \mathcal{N}(\mu^*(t) + \alpha(o - \mu^*(t)), \sigma_w^2)$;

(ii) $\mathcal{P}$ is static (proposition 5.1): $\hat{\mu}_t \equiv \hat{\mu}_0$ and $\mathcal{A}_t(\mathcal{P}) \equiv \hat{\mu}_0$;

(iii) the exogenous drift is bounded: $|\zeta| < \alpha \cdot |\hat{\mu}_0 - \mu_0|$.

In the coercive regime, the system's stale actions continuously pull μ^* toward the frozen estimate $\hat{\mu}_0$: $\mu^*(t+1) = \mu^*(t) + \zeta + \alpha(\hat{\mu}_0 - \mu^*(t))$, which is a mean-reverting process with equilibrium $\hat{\mu}_0 + \zeta/\alpha$. When $|\zeta/\alpha|$ is small, μ^* is held near $\hat{\mu}_0$, suppressing V_P and making the system appear healthy under the standard score.

Corollary 8.2 (Coercive Masking)**.** *In the coercive regime (definition 8.1), the standard score $\sigma_P(t) = 1/(1+V_P(t))$ can fail to detect FM-1: $\sigma_P(t) \approx 1$ for all $t \leq t_{\text{mask}}$, where $t_{\text{mask}} = O(\alpha^{-1} \log(\alpha/\zeta))$ is the* coercive horizon*. Beyond t_{mask}, exogenous drift overwhelms the coercive capacity and V_P diverges.*

Proof sketch. Under condition (iii) of definition 8.1, the gap $\delta_t = \mu^*(t) - \hat{\mu}_0$ satisfies $\delta_{t+1} = (1-\alpha)\delta_t + \zeta$. The fixed point is $\delta^* = \zeta/\alpha$, so $V_P(t) \leq \frac{1}{2}(\zeta/\alpha)^2 + \sigma_w^2/(2\alpha)$ in expectation, masking FM-1 while the coercive condition holds. Once the drift accumulates beyond the available control authority, the frozen model can no longer hold the environment in place and the underlying mismatch reappears. $\square$

Remark 8.3 (Compression by adaptation vs. compression by forcing)*.* From the algorithmic-regulation viewpoint, coercive masking matters because it can improve realized compressibility without improving the model. The external trajectory becomes simpler, but for the wrong reason: the controller is forcing the world into alignment rather than updating its internal state to match a changing world.

Definition 8.4 (Effort-Adjusted Coherence Index)**.** The *effort signal* at time t is any monotone proxy for the corrective actuation burden required to keep the world aligned with the system's present output. A generic observable surrogate is

$$E_t^{\text{obs}} := |\mathcal{A}_t(\mathcal{P}) - \mathcal{A}_{t-1}(\mathcal{P})|,$$

the magnitude of change in the system's actuation. In controlled synthetic settings, a sharper proxy is the realized coercive effort

$$E_t^{\text{sim}} := \alpha\, |a_t - s_t^{\text{free}}|,$$

where s_t^{free} is the state that would have arisen before applying control.

The *Effort-Adjusted Coherence Index* is

$$\mathcal{C}^E(I(S,t), t) := \min\bigl(\sigma_P(t) \cdot e^{-\lambda E_t / E_0},\ \sigma_A(t),\ \sigma_\Phi(t)\bigr),$$

where $E_0 > 0$ is a reference effort scale and $\lambda > 0$ is a sensitivity parameter. $\mathcal{C}^E$ penalizes increasing actuation effort, correcting for action-induced confounding: it discounts apparent coherence when the environment is being held in place rather than genuinely estimated. In action-effective settings where coercive masking is plausible, $\mathcal{C}^E$ is the primary closed-loop diagnostic and $\mathcal{C}$ is the corresponding apparent-fit baseline.

Definition 8.5 (Masking Gap)**.** The *masking gap* at time t is

$$M_t \;\coloneqq\; \mathcal{C}(I(S,t),t) - \mathcal{C}^E(I(S,t),t) \;\geq\; 0.$$

It quantifies how much apparent coherence is lost once corrective effort is charged explicitly. Large M_t indicates that the observed fit is being maintained by intervention rather than by an up-to-date model.

Remark 8.6 (Calibration guidance for E_t, E_0, and λ)*.* The effort proxy should follow the strongest observable hierarchy available: E_t^{sim} in controlled simulators, E_t^{obs} from direct actuator burden in deployed systems, and E_t^{log} from logged exposure or policy divergence when only observational traces are available.

The reference scale E_0 should be set to the typical healthy effort level of the domain, not to a global constant. In practice this means using a nominal controller effort in simulation or the median healthy effort over a temporally held-out prefix in logged data. In the particle tracker we use the prior-scale effort floor, in KuaiRand we use the median healthy KL divergence from the random-exposure baseline, and in deployed control systems the right default is the nominal actuator burden under steady operation.

The penalty λ controls how aggressively apparent coherence is discounted. Across the particle masking grid and the KuaiRand sensitivity sweep, values in the range $\lambda \in [2,3]$ provide the most stable trade-off, with $\lambda = 3$ serving as the default: $\lambda < 1$ under-penalizes masking, while $\lambda = 4$ tends to be more aggressive than the added separation usually justifies.

These sweeps function as the current ablations for proxy strength and effort sensitivity: they show that the main conclusions are stable across a useful operating range, while still making clear that calibration is part of the diagnostic rather than a nuisance parameter hidden from the analysis.

The empirical evidence for coercive masking comes from the particle-tracker experiment (section 7.7). In the passive FM-1 regime, the standard score and $\mathcal{C}^E$ coincide because no realized forcing of the world occurs, so $M_t = 0$. Under moderate coupling ($\alpha = 0.3$, $\lambda = 3.0$), the same frozen model reaches $\mathcal{C} = 0.949 \pm 0.012$ while $\mathcal{C}^E = 0.697 \pm 0.046$, so the mean masking gap is $M_t \approx 0.252$, showing that the apparent coherence is being purchased by intervention rather than adaptation.

9 Related Work

Tracking under non-stationarity. The paper sits at the intersection of three literatures that are usually kept apart: concept drift detection, finite-memory tracking, and action-coupled prediction. In the concept-drift literature, classical detectors such as ADWIN, DDM, and Page-Hinkley identify changepoints or sustained mean shifts in a monitored signal [Gama et al., 2014, Bifet and Gavaldà, 2007, Gama et al., 2004, Harries, 1999]. Those methods are the right passive baselines, and our results do not contest that: on passive streams they remain the stronger default alarms. The role of the Coherence Index (CI) is narrower. It is a structural diagnostic tied to model fidelity, actuation reproducibility, and update quality, not a universal replacement for generic changepoint detection.

The second neighborhood is nonstationary optimization, adaptive filtering, and online least-squares. There the same cube-root tension between variation and memory already appears, but in Euclidean loss, bandit, or regret-based settings rather than in a distributional W_2/Sinkhorn geometry [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. The lower-bound layer added here sharpens that connection without collapsing the two problems into one. In our setting, the cube-root law is exact for causal averaging estimators and reappears as a minimax obstruction for arbitrary window-restricted estimators on a ζ-Lipschitz drift class in the critical-window regime. Classical smoothers such as EWMA and Kalman filtering belong to the same broad family of variance-versus-responsiveness trade-offs, but they are typically studied as estimators rather than as components of a closed-loop diagnostic argument. Our EWMA results are included precisely to show that the cube-root law is a finite-memory phenomenon rather than a sliding-window artifact.

Transport-based diagnostics. WATCH uses Wasserstein geometry to detect distribution shift in feature space [Faber et al., 2022b,a]. Our work is adjacent, but the use of transport is different in function. There, Wasserstein distance is the detector. Here, W_2 and its debiased Sinkhorn surrogate are built into the tracking model itself: they quantify representation error, induce the lag–variance trade-off, and anchor the coherence diagnostics. The point is not simply that OT is a good shift statistic, but that in the present setting it is part of the geometry of what it means for a finite-memory tracker to stay current.

Filtering and closed-loop control. The representation–action–adaptation decomposition is motivated by standard state-estimation and control logic, not by a claim that every adaptive system must have three roles. ISS theory supplies the stability template [Sontag, 1989, Dashkovskiy et al., 2010, Mironchenko and Prieur, 2020], while classical filtering provides the concrete Gaussian realization used here [Jazwinski, 1970]. Once actions affect future observations, state update must solve a credit-assignment problem rather than a passive filtering problem.

Belief update semantics. The distinction between revision and update in dynamic worlds [Katsuno and Mendelzon, 1992] gives a formal language for the same issue. Bonanno [2024] makes explicit that action-conditioned belief update must distinguish exogenous change from state changes caused by the agent itself. We use this observation only to justify the presence of the action argument in the closed-loop update.

Algorithmic regulation and compression. Ruffini [2026] interprets good regulation through algorithmic compression of realized trajectories. This work adds a temporal constraint absent from that framework: even a useful compressed model becomes stale under drift. The cube-root law therefore complements the algorithmic-regulation viewpoint by quantifying how finite memory limits the duration for which compression remains valid in a changing environment.

Action-coupled prediction. The coercive masking regime is related to broader work on systems that change the data-generating process they seek to model. The claim is narrower than a general performative-prediction analysis: we isolate a diagnostic failure mode in which low observed error can be sustained by intervention, and we propose an effort-aware correction for that case. This connects most directly to performative prediction and feedback-loop analyses in recommendation and dataset construction, including [Perdomo et al., 2020, Hardt and Mendler-D"unner, 2023, Taori and Hashimoto, 2023, Pagan et al., 2023, Krauth et al., 2022, Eilat and Rosenfeld, 2023]. The main distinction is that we are not trying to solve the full causal problem of performative feedback; we are proposing a diagnostic for when intervention effort masks stale models in active or logged systems. In that sense, the paper is best read not as a full theory of performativity, but as an attempt to mark a specific epistemic boundary: when low error should stop counting as evidence of good adaptation.

10 Limitations and Open Directions

Finite-memory scope. The cube-root law of proposition 6.12 is exact for the finite-memory averaging class analyzed here, and proposition 6.16 extends the lower-bound side to arbitrary measurable estimators that only see the last m samples of a ζ-Lipschitz drift path. The cube-root exponent therefore survives beyond linear averaging in the critical-window regime, but this still falls short of a universal minimax result over arbitrary recursive or extrapolative adaptive estimators.

Geometry and observability. The present formulation uses W_2 and its debiased Sinkhorn approximation to measure model drift. Extending the same trade-off cleanly to other geometries, such as f-divergences or kernel MMD, remains open. Likewise, $\mathcal{C}^E$ (the Effort-Adjusted Coherence Index, CI^E) requires an observable proxy for intervention effort, and the preferred proxy is domain-dependent. This is especially relevant in performative or feedback-loop settings, where intervention and exposure are partially confounded [Perdomo et al., 2020, Hardt and Mendler-D"unner, 2023, Taori and Hashimoto, 2023, Krauth et al., 2022].

Benchmark dependence. The passive benchmarks use Page-Hinkley events as the operational reference for lead time, and different failure definitions would change the resulting alarm statistics. On active and logged benchmarks, the usefulness of $\mathcal{C}^E$ still depends on calibration choices such as the effort penalty λ, the warning threshold, and the local drift estimator $\hat{\zeta}_t$.

Effort proxies and σ_A surrogates. The practical hierarchy of effort proxies is strongest in simulation, weaker in direct observation, and weakest in logged recommendation data. Likewise, σ_A is exact only in settings with replicated or fully specified action channels. In the Gaussian tracker it is constructive, in the particle tracker it is a controlled surrogate based on actuation gap, and in KuaiRand it is a normalized exposure-entropy proxy rather than a direct estimate of $e^{-H(\mathcal{A}|\mathcal{P})}$. The logged benchmark should therefore be read as operational evidence for effort-aware diagnostics, not as full causal identification of intervention dynamics.

Scaling beyond present validation. The Gaussian and particle trackers validate the theory in controlled Gaussian and non-Gaussian settings, and the real-data benchmarks show that the diagnostics remain useful outside the synthetic environment. High-dimensional state spaces, richer action channels, and direct measurements of intervention effort are the next natural tests of the framework.

Opaque adaptation maps. For black-box Φ_t, the present manuscript provides only the template requirement that an ISS-Lyapunov functional V_Φ exist. The CEGIS-based recovery idea mentioned in remark 6.18 is not implemented or validated in this repository, so diagnosing σ_Φ for opaque adaptation subsystems remains an open engineering problem.

Computational profile. The Sinkhorn runtime study is a prototype-level NumPy benchmark on Gaussian clouds. It is useful for separating statistical accuracy from wall-clock cost, but it does not claim a production implementation ceiling. The main practical message is that regularization ε and window size dominate the cost–bias trade-off, while higher-dimensional settings mainly affect estimator bias under tight regularization.

Missing adaptive baselines. The passive comparisons emphasize ADWIN, Page-Hinkley, KSWIN, and the NoDrift sanity baseline, and now also include CUSUM, forgetting-factor recursive least squares, and scalar Kalman predictors on the real-world streams. That does not exhaust the baseline space: additional representation-space or multivariate passive detectors remain future work, and the present calibration should still be read as benchmark-specific rather than universal.

11 Conclusion

The paper argues for a simple but consequential view of adaptive systems. Under drift, memory is not only a resource; it is also a source of staleness. Under feedback, good observed fit is not yet evidence that the model is current, because intervention can keep the world near a stale estimate. The two claims of the paper follow from taking those facts seriously.

On the tracking side, the finite-memory averaging class obeys a sharp coherence-delay limit. When total error is written as a sum of estimation variance and drift-induced staleness, optimizing over memory yields the cube-root law $\mathcal{E}_{\min} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$ for the estimator class analyzed here. The broader lower-bound argument then shows that the same exponent is not merely a sliding-window artifact: it reappears for arbitrary estimators restricted to the last m samples on a ζ-Lipschitz drift class in the critical-window regime. The practical lesson is direct. More data can still make a tracker worse if the world moves faster than the memory can stay current.

On the diagnostic side, action-coupled systems admit a specific failure mode in which apparent coherence is maintained by forcing rather than by adaptation. The decomposition into representation, actuation, and adaptation roles $(\mathcal{P}, \mathcal{A}, \Phi)$ isolates that regime from ordinary degradation, and the Effort-Adjusted Coherence Index (CI^E) is designed precisely for it. The empirical results support that split. The Gaussian tracker recovers the expected U-curve and the $\zeta^{1/3}$ scaling law for both sliding windows and EWMA. The particle benchmark shows that feedback masking can keep the standard score high even when the model is stale. On passive streams, generic changepoint detectors such as ADWIN remain the stronger default alarms. On active and logged benchmarks, however, the effort-aware score becomes the right diagnostic because the failure mode is no longer passive drift alone.

The broader claim is therefore modest but, we think, useful. Adaptive systems should not be evaluated only by how well they fit what they currently see. They should also be evaluated by how much temporal staleness they can tolerate and by how much intervention is required to preserve the appearance of coherence. Within the scope studied here, that is the main lesson: finite memory imposes a real tracking floor, and under feedback, raw fit is not yet a safe measure of adaptation.

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A Code Repository

All implementations are publicly available at:

https://github.com/jovemexausto/coherence-delay-tradeoff

The repository contains:

- `experiments/run.py` — single entry point for all experiment domains;
- `experiments/gaussian/` and `experiments/particle/` — core experiment logic;
- `experiments/bikes/`, `experiments/elec2/`, and `experiments/kuairand/` — domain-specific benchmarks;
- `REPRODUCIBILITY.md` — setup and run instructions.

All experiments use `uv` for dependency management (`uv sync`) and are reproducible with fixed random seeds as specified in each script.

B KuaiRand Logged Benchmark Protocol

This protocol instantiates the real-data masking benchmark on top of the KuaiRand sequential recommendation logs. The goal is to test whether the Effort-Adjusted Coherence Index (CI^E) can surface exposure concentration and filter-bubble-like drift better than passive detectors and a smoothed-score baseline.

Data and signals

The primary state signal is `watch_ratio`. Auxiliary feedback signals include `click` and `like`; `dwell`, `follow`, and `share` are reserved for secondary validation when available. For user u at time t, let $P_{u,t}$ denote the inferred preference state, $A_{u,t}$ the exposure policy, and $\Phi_{u,t}$ the composite feedback signal. All user windows use `window_size = 20`, and video tags are binned from `video_features_basic_pure.csv`. The benchmark keeps only users with at least 20 interactions in the healthy, bubble, and collapse segments, yielding the 367-user evaluation cohort used in the main text.

Operational score components

In KuaiRand, σ_P is a normalized deviation of the current watch-ratio window from the healthy watch-ratio baseline, σ_Φ is a stability term based on within-window watch-ratio dispersion, and σ_A is an operational surrogate based on normalized exposure-tag entropy. This last term should be read as a logged-data proxy for action reproducibility, not as a direct estimator of $e^{-H(\mathcal{A}|\mathcal{P})}$, because the benchmark has no replicated policy logs.

The proxy hierarchy matters: if direct action logs or replica logs exist, the Gaussian-style exact interpretation can be approximated; if not, the logged-data entropy surrogate is the intended operational choice. We keep the report honest by treating the logged benchmark as evidence for a useful proxy, not as a proof that the surrogate equals the unobserved conditional entropy. In the ladder used throughout the paper, KuaiRand is a proxy-level benchmark.

Effort proxy

The effort proxy is defined as the divergence between the current exposure distribution and the healthy random-exposure baseline distribution:

$$E_{u,t} := D_{\mathrm{KL}}\left(\pi_{u,t}^{\text{current}} \,\middle\|\, \pi_{u,t}^{\text{healthy}}\right).$$

This choice is directional, decomposable, and directly interpretable as the cost of keeping the observed trajectory close to a healthy baseline. In settings where only aggregate exposure counts are available, total variation distance may be used as a fallback, but KL is the preferred default. We use additive smoothing 10^{-6} in the KL computation.

Protocol

The benchmark is executed in three phases.

1. **Healthy.** Exposure follows the random baseline, diversity is unconstrained, and effort remains low.
2. **Bubble.** Exposure becomes progressively concentrated, so that $E_{u,t}$ rises while standard coherence may remain artificially high.
3. **Collapse.** The concentrated exposure regime stops tracking the healthy baseline, generic detectors begin to fire, and the effort-corrected index should lead the baselines.

Evaluation

Evaluation is carried out sequentially within each user session and then batched across users with strict temporal splits. Thresholds for $\mathcal{C}$, $\mathcal{C}^E$, and EWMA-smoothed $\mathcal{C}^E$ are calibrated on healthy windows only, using the 20th percentile of the healthy score distribution. The reference effort scale E_0 is the median healthy KL divergence. Reported metrics are:

- detection rate of the bubble/collapse phases,
- median detection delay,
- false positives in the healthy phase,
- sensitivity to λ in $\mathcal{C}^E$,
- comparison against ADWIN, PageHinkley, KSWIN, NoDrift, and a smoothed CI^E baseline (EWMA).

All baseline detectors consume the same effort-adjusted scalar signal, $1-\mathcal{C}^E$, so the comparison is not biased by detector-specific inputs. The baseline hyperparameters are fixed a priori: ADWIN $\delta = 0.03$; PageHinkley $\delta = 0.005$, threshold $= 20$, $\alpha = 0.9999$; KSWIN window $= 30$, stat $= 10$, $\alpha = 0.001$. When enabled, thresholds are calibrated on healthy windows only.

Controls

Required controls include shuffled labels, random phase boundaries, no- coercion ablations, and removal of auxiliary feedback signals. These controls ensure that the benchmark does not reduce to a hindsight-tuned logged-bandit exercise.